

Design of Silicon Photonic Multimode Interference Couplers

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RSoft Design is a software used by researchers, system integrators, manufacturers and service providers, thanks to its tools which provide a wide range of applications, from the design of photonic components to planning and design of systems and networks. Regarding the design of photonic components it is possible to model active and passive components, for both purposes there is a Device Suit, and each one includes a CAD environment, simulation engines with one with a different calculation method, and an optimization utility.

The targeted photonic component to be designed was the MMI Coupler, that follows the self-imaging principle of multimodal waveguides, the application of this kind of devices has grown due to the interest of its effects in integrated optics and its advantages, this component specifically offers higher fabrication tolerances and polarization independent results, both features of great convenience regarding the different applications of this component in integrated circuits.

The MMI Coupler falls into the category of a passive photonic component, and from the series of engines which could be applied, the BeamPROP engine, based on the Beam Propagation Method, is implemented for its accuracy in the final results of the simulations, and its outstanding computational performance.

The design of the Coupler was based on the Fraunhofer IZM model for waveguides, which is a SOI structure with specific standard dimensions, and as simulations were performed particular geometric characteristics, were implemented to obtain an optimized result. The geometric characteristics mentioned before are the choosing of an appropriate width for the MMI section which resulted in the best possible output, the tapering of the access segments between the MMI section and the monomode waveguides to create a better balance between the different outputs and to obtain a higher power in the same point, and finally, when necessary the tapering of the MMI section to avoid cross-talk between the monomode segments after the coupling section.

RSoft Design es un software usado por investigadores, integradores de sistemas, fabricantes y proveedores de servicios, gracias a sus herramientas que ofrecen un rango considerablemente amplio de aplicaciones, desde el diseño de componentes fotónicos hasta la planificación y diseño de sistemas y redes. En lo referente a el diseño de componentes fotónicos es posible modelar componentes activos y componentes pasivos, y en ambos casos se puede contar con un Device Suit, un medio CAD, motores de simulación, cada uno con diversos métodos de cálculo, y utilidades de optimización.

El componente fotónico a ser diseñado fue el acoplador de Interferencia Multimodal, que sigue los principios de autoimagen característicos de las guías de onda multimodal, la aplicación de estos dispositivos ha aumentado enormemente debido al interés de sus efectos en las ópticas integradas y sus ventajas, este componente ofrece específicamente una tolerancia de fabricación considerablemente elevada y resultados que son independientes de la polarización usada, ambas características convenientes en la aplicación de éste componente en circuitos integrados.

El acoplador antes mencionado se encuentra categorizado como un componente fotónico pasivo, y de la serie de motores utilizables para estos modelos, el motor BeamPROP, basado en el Método de Propagación del Haz, es implementado, por su precisión en los resultados finales de las simulaciones y por el excepcional rendimiento computacional que presenta.

El diseño del acoplador se basó en el modelo de guía de ondas de Fraunhofer IZM, el cual es una estructura SOI con dimensiones estándares específicos, y al llevar a cabo numerosas simulaciones fueron introducidas ciertas particularidades geométricas para obtener resultados optimizados. Las características geométricas mencionadas previamente fueron, la escogencia de una anchura apropiada para la sección MMI que produjese el mejor resultado posible, la disminución o aumento gradual de los segmentos de entrada, entre la sección MMI y las guías de onda monomodo para así crear un mejor balance entre las diferentes salidas y así obtener en el mismo punto una potencia más elevada, y finalmente, cuando fuese necesario la disminución o aumento gradual de la sección MMI para evitar diafonía entre los segmentos monomodo luego del sector de acoplamiento.

RSoft Design és un programari usat per investigadors, integradors de sistemes, fabricants i proveïdors de serveis, gràcies a les seves eines que ofereixen un rang considerablement ampli d'aplicacions, des del disseny de components fotònics fins a la planificació i disseny de sistemes i xarxes. Pel que fa a el disseny de components fotònics és possible modelar components actius i components passius, i en ambdós casos es pot comptar amb un Device Suit, un mitjà CAD, motors de simulació, cadascun amb diversos mètodes de càlcul, i utilitats d'optimització.

El component fotònic a ser dissenyat va ser el acoblador d'Interferència Multimodal, que segueix els principis de autoimatge característics de les guies d'ona multimodal, l'aplicació d'aquests dispositius ha augmentat enormement causa de l'interès dels seus efectes en les òptiques integrades i les seves avantatges, aquest component ofereix específicament una tolerància de fabricació considerablement elevada i resultats que són independents de la polarització utilitzada, ambdues característiques convenients en l'aplicació d'aquest component en circuits integrats.

L'acoblador abans esmentat està categoritzat com un component fotònic passiu, i de la sèrie de motors utilitzables per aquests models, el motor BeamPROP, basat en el Mètode de Propagació del Feix, és implementat, pel seu precisió en els resultats nals de les simulacions i per l'excel·lent rendiment computacional que presenta.

El disseny del acoblador es va basar en el model de guia d'ones de Fraunhofer IZM, el qual és una estructura SOI amb dimensions estàndards específiques, i en dur a terme nombroses simulacions van ser introduïdes certes particularitats geomètriques per obtenir resultats optimitzats. Les característiques geomètriques esmentades prèviament van ser la tria d'una amplada apropiada per a la secció MMI que produís el millor resultat possible, la disminució o augment gradual dels segments d'entrada, entre la secció MMI i les guies d'ona monomode per així crear un millor balanç entre les diferents sortides i així obtenir en el mateix punt una potència més elevada, i finalment, quan fos necessari la disminució o augment gradual de la secció MMI per evitar diafonia entre els segments monomode després del sector d'acoblament.

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List of Acronyms

2D	Two-Dimensional
3D	Three-Dimensional
AC	Alternating Current
AWG	Arrayed Waveguide Grating
BESOI	Bond and Etch-back SOI
BPM	Beam Propagation Method
CAD	Computer Aided Design
CAMFR	CAvity Modelling FRamework
CD	Critical Dimension
CMOS	Complementary Metal Oxide Semiconductor
CMP	Chemical Mechanical Polishing
CMT	Coupled Mode Theory
CVD	Chemical Vapor Deposition
Cz	Czochalski
DC	Direct Current
DOE	Diffraction Optical Elements
EL	Excess Loss
EM	Electro-Magnetic

EMEM EigenMode Expansion Method

EIM Effective Index Method

FD-BPM Finite-Difference Beam Propagation Method

FDTD Finite-Difference Time-Domain

FDM Finite-Difference Method

FE-BPM Finite-Element Beam Propagation Method

FEM Finite Element Method

FFT-BPM Fast Fourier Transformer Beam Propagation Method

FIB Focused Ion Beam

FT Fabrication Tolerance

FV-FDM Full-Vectorial Finite-Difference Method

FZ Floating Zone

IF Intermediate Frequency

IL Insertion Loss

IZM Institut für Zuverlässigkeit und Mikrointegration

LED Light Emitting Diodes

LO Local Oscillator

MBE Molecular Beam Epitaxy

MMI MultiModal Interference

MPA Modal Propagation Analysis

MTL Modal Transmission Line

MZI Mach-Zehnder Interferometer

PC Photonic Crystal

PDE Partial Differential Equations

PDL Polarization-Dependent Loss

PICs Photonic Integrated Circuits

PLC Planar Light-wave Circuits

PSR Power Splitting Ratio

PWE Plane Wave Expansion

RCWA Rigorous Coupled Wave Analysis

RIE Reactive Ion Etch

SIM Spectral Index Method

SIMOX Separation by IMplanted OXygen

SOI Silicon On Insulator

SPC Statistical Process Control

SSM Split-Step Method

TE Transverse Electric

TEM Transverse Electro-Magnetic

TIR Total internal reflection

TM Transverse Magnetic

TMI Two Mode Interference

TMM Transfer Matrix Method

VFE-BPM Full-Vector Finite-Element Beam Propagation Method

WDM Wavelength-Division Multiplexing

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Chapter 1

Introduction

In modern days telecommunication networks are having a enormous growth, creating several needs for them to continue providing quality services. Reconfigurability, flexibility and speed, are some of the most important of the mentioned needs, from this search optical communications flourished solving many problems by the use of light as the information transmission medium. These information signals must be in the visible spectrum, or in the infrared spectrum (850 nm to 1650 nm).

Light is nor considered a wave or a particle but both for this reason it is defined as photonic. The science of photonics [13] refers to the manipulation of light, whether it is the generation, emission, transmission, modulation, signal processing, switching, amplification, or detection/sensing.

To manipulate the light it is required the use of dielectric material waveguides with high permittivity, and thus high index of refraction to create a total internal reflection guiding the light as needed. Also, the use of Photonic Integrated Circuits (PICs) is required, these are devices that integrate several functions for optical communications to be possible.

The use of Silicon in this technologies is a consequence of its success in microelectronics and the search of a platform to achieve a monolithic integration of optics and microelectronics. Silicon for optical interconnects brought many challenges [14] like the high propagation losses (due to scattering off the waveguide's sidewalls), the low electro-optic coefficient, the low light-emission efficiency, and high fiber-to-waveguide coupling losses. All this was overcome with the use of new nanofabrication techniques, enabling the demonstration of a large number of ultracompact high performance photonic components.

Silicon Photonics [15] which is defined as the study and application of photonic systems which use silicon as an optical medium.

Couplers have proven to be essential components representing the biggest market of photonic integrated circuits, and find use in broadcast-type optical networks and for optical signal routing and processing.

In this project we have designed the MultiModal Interference (MMI) Coupler on Silicon On Insulator (SOI), for its advantages, which were enhanced by the use of tapered structures and dimension optimization, diminishing losses and eliminating cross-talk at the outputs. The results obtained show optimized and uniform outputs for several MMI coupler structures, varying the number of inputs and outputs like the 2×2 structure which is of great importance in the Mach-Zehnder Interferometer (MZI) and the 1×3 structure which by having more outputs requires certain geometric modifications. Fabrication tolerances are also reported for future possible industrial use.

1.1 Background and Scope

1.1.1 Background

The most common of couplers is the Directional Coupler which produces devices with large dimensions, specially for a high number of outputs, and also it presents low fabrication tolerances. These components are currently used in Fraunhofer Institut für Zuverlässigkeit und Mikrointegration (IZM) and its disadvantages are the main reason for the search of more suitable solutions.

The MMI Couplers based on the principle of self imaging solves all these problems with properties such as compactness, high fabrication tolerance, inherent output power balance, polarization independence and low optical loss. Such number of advantages makes it clear that this coupler is the best option in the fabrication of more elaborated optical circuits.

This passive components are employed as power splitters and combiners in MZI and optical switches, and in many other applications, creating a increasing popularity in its use for integrated optical circuits.

1.1.2 Related Work

The design of MMI Couplers has taken place in many scenarios, varying the wavelength used for the light transmission depending on the absorption of each material, the material systems of the structure including $LiNbO_3$, Al_2O_3/SiO_2 on Si , $InGaAsP/InP$ and $GaAs/AlGaAs$. The most common is Si on SiO_2 , called SOI, also used in this project with the commonly applied wavelength of $1,55\mu m$.

Many, as well, have found appealing the use of MMI Couplers with changing widths, tapered MMI sections, with parabolical and exponential change. Also, the use of a linear tapering segmente, between the monomode segments and the MMI section, is a common solution to large losses.

1.1.3 Scope

This project is limited to the use of computational analysis, by the use of the Simulation tool Rsoft with the BeamPROP engine, which functions with the Beam Propagation Method (BPM). Aldo,

in several points it is necessary the use of numerical methods to determine the mathematical background that led to certain decisions and others that verify the results obtained with the method mentioned at first.

Also, the structures displayed and tested are only with a low number of outputs since the MMI Couplers are specially suitable for this use. The material system approached is limited to the SOI, and the wavelength to be studied would only be the most common one in photonics $1,55\mu m$.

1.2 Objectives

The main objective of this project is to find a feasible MMI Coupler that will generate a compact and fabrication tolerant MMI section, with a low loss and balanced output, so its can be applicable in PICs working efficiently and properly. A more schematic way to reach this goal is specified in the following steps:

1. The definition of the most suitable MMI dimensions.
2. Research and find possible third party solutions for its optimization.
3. Implementation of a new solution.
4. Perform simulations and evaluate the most efficient of this solutions.

1.3 Chapter Overview

This project is structured in the following way: the next Chapters, 2 and 3 , contain basic information related to the component in question. Chapter 2 exposes the details of optical waveguides and the theory behind it. Chapter 3 explains coupling mode theory detailing its characteristics and equations for the different mode polarizations.

Chapter 4 is centered in general theoretical backgrounds related to the self imaging principle that is the basis to the behavior of the MMI devices, and therefore to coupler in question. Chapter 5 is related specifically to the fabrication methods used to create this devices.

Chapter 6 views the diversity in design tools and analysis, explaining all the numerical methods that are valid in our quest, it is also focused extensively on the design and simulation steps taken to obtain the final product of this project, and Chapter 7 is focused on several applications of the MMI Coupler, detailing its role in each one. Finally, Chapter 8 contains the conclusions and future prospects.

Chapter 2

Optical Waveguides

The basic concepts and equations of Electro-Magnetic (EM) wave theory required for the understanding of light propagation in optical waveguides. Light confinement and mode formations in the waveguide are explained in detail. Maxwell equations and boundary conditions are also a point of focus. And finally, the characteristics of polarization dependence, particularly birefringence, and general considerations in SOI waveguides.

2.1 Waveguide Concepts

Optical waveguides are physical dielectric structures that are able to transport EM energy. Optical fibers and optical waveguides [1] is characterized by a central region, called the core, where the light is confined, it is composed by a refractive index(n_1) which is higher than that of the surrounding region of normally uniform index(n_0), this second region is called the cladding, all this structure is shown in the Figure 2.1 .

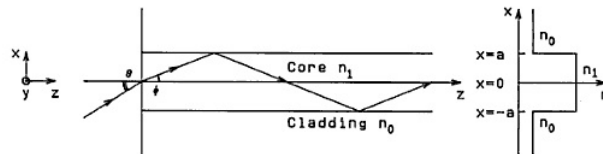


Figure 2.1: Basic Structure and refractive index profile of the optical waveguide [1]

2.1.1 Waveguide Parameters

The propagation characteristics of a waveguide can be expressed in terms of certain parameters, this parameters are explained in the following subsection and shown in Figure 2.1 , which is a typical waveguide cross-section. Also, when necessary, equations to describe and obtain this parameters will be displayed.

2.1.1.1 Refractive Index

In optics the refractive index [16] , n , of a substance (optical medium) is a number that describes how light, or any other radiation, propagates through that medium.

Its most elementary occurrence (and historically the first one) is in Snell's law of refraction, $n_1 \sin \phi_1 = n_0 \sin \phi_0$, where θ_1 and θ_0 are the angles of incidence of a ray crossing the interface between two media with refractive indices n_1 and n_0 . Brewster's angle, the critical angle for total internal reflection, and the reflectivity of a surface also depend on the refractive index, as described by the Fresnel equations.

2.1.1.2 Birefringence

The birefringence [17] or strength of the double refraction is the property of having two refractive indices, and the numerical difference between the minimum and maximum refractive indices is its quantifiable value (Equation 2.1), this difference in refractive indices occurs in the directions parallel and perpendicular to the direction of orientation, it depends on the polarization and propagation direction of light. These materials are said to be optically anisotropic, meaning they have different properties in different directions. Another definition is the measure of the total molecular orientation. This effect was first described by the Danish scientist Rasmus Bartholin in 1669.

$$\Delta \eta = \eta_{\parallel} - \eta_{\perp} \quad (2.1)$$

2.1.1.3 Total Internal Reflection

Basically, the guiding of the light is a consequence of a total internal reflection from the interface between the core and the cladding, generated by the fact that the refractive index of the core n_1 is higher than the index of the cladding. The condition for total internal reflection is given by Equation 2.2.

$$n_1 \sin\left(\frac{\pi}{2} - \phi\right) \quad (2.2)$$

For the Total Internal Reflection [2] to occur without any major losses, the angle of incident light must be bigger than the critical angle $\theta_c = \sin\left(\frac{n_2}{n_1}\right)$, as shown in the Figure 2.2.

Where n_1 is the index of the core and n_2 the index of the cladding. Although, in this Figure we can conclude that after the condition is followed there is no energy loss, but in reality small losses exist due to the absorption in the medium and to the reflections in the surfaces where the light enters and leaves the medium.

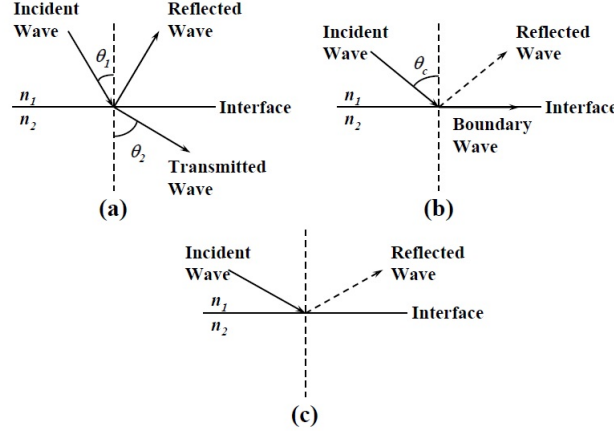


Figure 2.2: Reflection and refraction of a parallel beam [2]

2.1.1.4 Relative Refractive-Index Difference

Relative Refractive-Index Difference is defined as a percentage that represents the difference between n_1 and n_0 . It is defined by the Equation 2.3.

$$\Delta = \frac{n_1^2 - n_0^2}{2n_1^2} \cong \frac{n_1 - n_0}{n_1} \quad (2.3)$$

2.1.1.5 Numerical Aperture

Numerical Aperture [2] is defined as the maximum angle of acceptance of a waveguide. Numerically we can obtain it by parting from Equation 2.2 of total internal reflection in Subsection 2.1.1.3, and since the angle ϕ is related to the incident angle θ by the Equation 2.4.

$$\sin \theta = n_1 \sin \phi \leq \sqrt{n_1^2 - n_0^2} \quad (2.4)$$

We can obtain an equation that represents the maximum angle of acceptance, resulting the shown in the next Equation.

$$\theta \leq \sin^{-1} \sqrt{n_1^2 - n_0^2} \equiv \theta_{max} \quad (2.5)$$

It can also be expressed in function of the Relative Refractive-Index Difference, as shown:

$$\theta_{max} = NA = n_1 \sqrt{2\Delta} \quad (2.6)$$

2.1.2 Guided Modes Formation

For the light to propagate through the waveguide the ray must have a specific discrete angle, which excites a particular mode, meaning, each discrete angle is associated to a mode [2]. Modes can also be described with the ray picture in the slab waveguide, as shown in Figure 2.3 where it is possible to observe the phase fronts which are perpendicular to the incident light rays, the wavelength and the wavenumber, a constant representing the average phase variation of the field ϕ , of light in the core are the following Equations.

$$\lambda = \frac{\lambda_0}{n_1} \quad (2.7)$$

$$k = k_0 n_1 \quad (2.8)$$

Where λ_0 is the wavelength in vacuum and $k_0 = \frac{2\pi}{\lambda_0}$. Therefore, the propagation constants, which are very important characteristics of the mode, in the lateral direction (x and z) are given by the next Equations.

$$\beta = k_0 n_1 \cos \phi \quad (2.9)$$

$$\kappa = k_0 n_1 \sin \phi \quad (2.10)$$

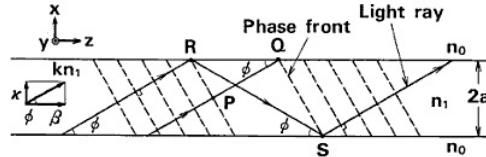


Figure 2.3: Picture of "modes" propagating along a waveguide [2]

There are various types of modes, in the classification by polarization we can encounter the Transverse Electric (TE) modes, characterized for having no electric field in the direction of propagation, and the Transverse Magnetic (TM) modes which are lacking the magnetic field in the direction of propagation. We can also have the Transverse Electro-Magnetic (TEM) modes that don't have neither electric nor magnetic fields in the direction of propagation, and finally the Hybrid modes, called this way for its non-zero electric and magnetic fields in the direction of propagation.

2.1.3 Maxwell's Equations

In a dielectric medium we can express with Maxwell's Equations [3] in the following form Equations.

$$\nabla \times E = -\mu \frac{\partial H}{\partial t} = ik \left(\frac{\mu_0}{\epsilon_0} \right)^{\frac{1}{2}} H \quad (2.11)$$

$$\nabla \times H = \epsilon \frac{\partial E}{\partial t} = -ikn^2 \left(\frac{\mu_0}{\epsilon_0} \right)^{\frac{1}{2}} E \quad (2.12)$$

Where μ and ϵ represent the permeability and permittivity of the medium, respectively. And can be related to their respective values in vacuum, $\mu_0[H/m]$ and $\epsilon_0[F/m]$.

$$\epsilon = \epsilon_0 n^2 \quad (2.13)$$

$$\mu = \mu_0 \quad (2.14)$$

We can establish electric and magnetic fields in function of the position in the plane transverse to the z -axis, r , considering that it has an angular frequency ω and it propagates in the z direction with propagation constant β . Obtaining:

$$e = E(r)e^{j(\omega t - \beta z)} \quad (2.15)$$

$$h = H(r)e^{j(\omega t - \beta z)} \quad (2.16)$$

With Equations 2.11, 2.12, 2.15 and 2.16 we can get in terms of Cartesian Coordinates the equations, which are the bases to waveguide analysis.

$$\left\{ \begin{array}{lcl} \frac{\partial E_z}{\partial y} + j\beta E_y & = & -j\omega\mu_0 H_x \\ -j\beta E_x - \frac{\partial E_z}{\partial x} & = & -j\omega\mu_0 H_y \\ \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} & = & -j\omega\mu_0 H_z \\ \frac{\partial H_z}{\partial y} + j\beta H_y & = & j\omega\epsilon_0 n^2 E_x \\ -j\beta H_x - \frac{\partial H_z}{\partial x} & = & j\omega\epsilon_0 n^2 E_y \\ \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} & = & j\omega\epsilon_0 n^2 E_z \end{array} \right. \quad (2.17)$$

2.2 Planar Waveguides

Planar Waveguides [citeokamoto2005] fundamentals are the basis of waveguide theory used in PICs and in semi-conductor lasers, its mathematical background is fit to be used in the development of Two-Dimensional (2D) slab waveguides and for the rectangular waveguide which requires a Three-Dimensional (3D) analysis, this section will explain and analyze the mentioned waveguides to obtain a full knowledge of optical waveguide theory.

2.2.1 Slab Waveguides

The slab waveguide [3] , shown in Figure 2.4 , it's the most basic of planar waveguides and it consists of a very simple geometry, the core is sandwiched between two layers of materials with lower refractive index. The lower layer is commonly a substrate with an index of n_s . The top layer is called the cover and its refractive index is n_c (in some bibliographies n_0) and can not be equal to the refractive index of the substrate (in symmetrical waveguides $n_c = n_s$). Usually the cover is Air which has a refractive index of $n_c = 1$.

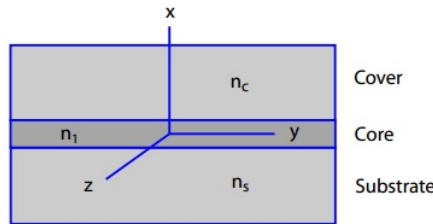


Figure 2.4: Slab Waveguide [3]

The propagation characteristics belonging to this wave analysis have already been explained in Section 2.2 , and therefore the Maxwell Equations are equal to the ones in Equations 2.11 and 2.12 . Also, the same EM field components defined in Equation 2.17 , with the difference that the slab waveguide as we know is a 2D structure and so the EM fields E and H lack of a $y - axis$ dependency setting the following values.

$$\frac{\partial E}{\partial y} = 0 \quad (2.18)$$

$$\frac{\partial H}{\partial y} = 0 \quad (2.19)$$

And all the field components vary with z , this occurs for all modes, thus:

$$\frac{\partial E}{\partial z} = j\beta E \quad (2.20)$$

$$\frac{\partial H}{\partial z} = j\beta H \quad (2.21)$$

These equations produce two different sets of linearly polarized solutions, meaning that we obtain two independent EM modes.

For the TE mode, name which is obtained since the electric field lies in the plane that is perpendicular to the z - *axis*, $E_z = 0$, $H_y = 0$ and $E_x = 0$, and we have the following equations that are satisfied:

$$\frac{d^2 E_y}{dx^2} + (k_0^2 n^2 - \beta^2) E_y = 0 \quad (2.22)$$

where

$$k_0 = \omega \sqrt{\epsilon_0 \mu_0} \quad (2.23)$$

$$H_x = -\frac{\beta}{\omega \mu_0} E_y \quad (2.24)$$

$$H_z = -\frac{j}{\omega \mu_0} \frac{dE_y}{dx} \quad (2.25)$$

For TM modes, name given by the fact that the magnetic field lies in the plane that is perpendicular to the z - *axis*, $H_z = 0$, $H_x = 0$ and $E_y = 0$. And are obtained by solving the same Equation 2.22 written as:

$$\frac{d}{dx} \left(\frac{1}{n^2} \frac{dH_y}{dx} \right) + (k^2 - \frac{\beta^2}{n^2}) H_y = 0 \quad (2.26)$$

where

$$E_x = -\frac{\beta}{\omega \epsilon_0 n^2} H_y \quad (2.27)$$

$$E_z = -\frac{j}{\omega \epsilon_0 n^2} \frac{dH_y}{dx} \quad (2.28)$$

Having the previous equations we can solve separately each layer of the slab waveguide to calculate the boundary conditions, obtaining:

$$\begin{cases} A \cos(\kappa a - \phi) \exp^{-\sigma(x-a)} & (x > a) \\ A \cos(\kappa x - \phi) & (-a \leq x \leq a) \\ A \cos(\kappa a + \phi) \exp^{\xi(x+a)} & (x < -a) \end{cases} \quad (2.29)$$

where κ , ξ and σ are wavenumbers in the x - *axis* in the core and cladding layers. They are defined as:

$$\begin{cases} \kappa^2 = k_0^2 n_1^2 - \beta^2 \\ \sigma^2 = \beta^2 - k_0^2 n_0^2 \\ \xi^2 = \beta^2 - k_0^2 n_s^2 \end{cases} \quad (2.30)$$

At the H_z boundaries the components are continuous, so based on the previous equations:

$$\frac{dE_y}{dx} = \begin{cases} -\sigma A \cos(\kappa a - \phi) \exp^{-\sigma(x-a)} & (x > a) \\ -\kappa A \cos(\kappa x - \phi) & (-a \leq x \leq a) \\ \xi A \cos(\kappa a + \phi) \exp^{\xi(x+a)} & (x < -a) \end{cases} \quad (2.31)$$

And the continuity in the boundaries gives:

$$\begin{cases} \sigma A \cos(\kappa a - \phi) = \kappa A \cos(\kappa x - \phi) \\ \kappa A \cos(\kappa x + \phi) = \xi A \cos(\kappa a + \phi) \end{cases} \quad (2.32)$$

If we substitute the following we can obtain the eigenvalues mentioned ahead in Equations 2.34 and 2.36 :

$$\begin{cases} u = \kappa a \\ w = \xi a \\ w' = \sigma a \end{cases} \quad (2.33)$$

$$u = \frac{m\pi}{2} + \frac{1}{2} \tan^{-1} \left(\frac{w}{u} \right) + \frac{1}{2} \tan^{-1} \left(\frac{w'}{u} \right) \quad (2.34)$$

$$\phi = \frac{m\pi}{2} + \frac{1}{2} \tan^{-1} \left(\frac{w}{u} \right) - \frac{1}{2} \tan^{-1} \left(\frac{w'}{u} \right) \quad (2.35)$$

Where $m = 0, 1, 2, \dots$

2.2.2 Rectangular Waveguides

The rectangular waveguide [3] has 3D characteristics and therefore must have another analysis. In these waveguides the light is confined in both the x and y dimensions. In the middle core layer the high index region has a finite width of $2w$ and is surrounded on all sides by lower index materials, the refractive index can be different in every side, as shown in Figure 2.5 .

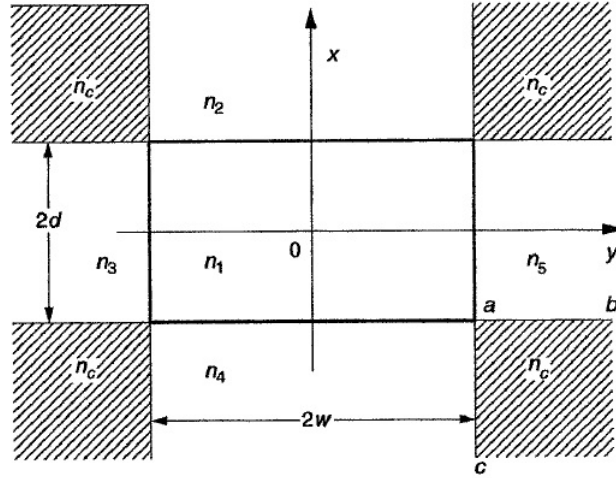


Figure 2.5: Rectangular Waveguide [3]

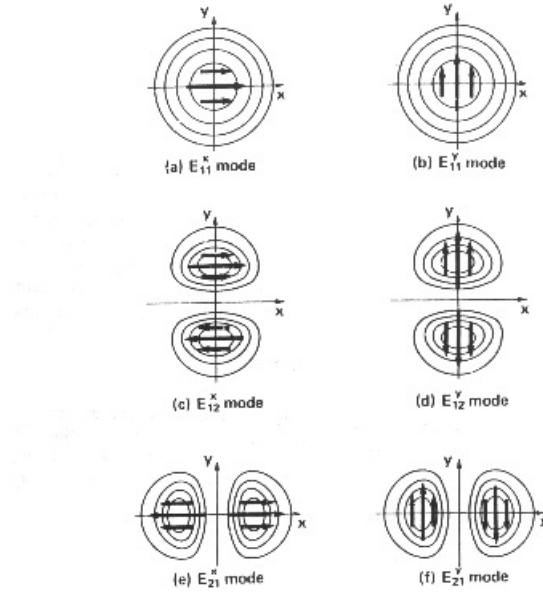


Figure 2.6: mode definitions and electric field distributions in Marcatili's method [3]

To analyze the best method to follow is the proposed by Marcatili which suggests 2 simplifications, to ignore the boundary conditions associated with hatched regions and to assume core-cladding index differences are small on all sides. In Figure 2.6 we can see the mode definitions and the electric field distribution (no fields in the corners of the waveguide).

Taking all these measures into account we can then simply proceed to analyze the rectangular waveguide as two slab waveguides, one in the y direction and another in the x direction.

First considering EM mode in which E_x and H_y are predominant and setting $H_x = 0$ we

obtain the wave equation and the EM field representation.

$$\frac{\partial^2 H_y}{\partial x^2} + \frac{\partial^2 H_y}{\partial y^2} + (k_0^2 n^2 - \beta^2) H_y = 0 \quad (2.36)$$

$$\begin{cases} E_x = \frac{\omega\mu_0}{\beta} H_y + \frac{1}{\omega\epsilon_0 n^2 \beta} \frac{\partial^2 H_y}{\partial x^2} \\ E_y = \frac{1}{\omega\epsilon_0 n^2 \beta} \frac{\partial^2 H_y}{\partial x \partial y} \\ E_z = \frac{-j}{\omega\epsilon_0 n^2} \frac{\partial H_y}{\partial x} \\ H_z = \frac{-j}{\beta} \frac{\partial H_y}{\partial y} \end{cases} \quad (2.37)$$

For the other dimension we set $H_y = 0$ and consider the EM field in which E_y and H_x are predominant we obtain:

$$\frac{\partial^2 H_x}{\partial x^2} + \frac{\partial^2 H_x}{\partial y^2} + (k_0^2 n^2 - \beta^2) H_x = 0 \quad (2.38)$$

$$\begin{cases} E_x = -\frac{1}{\omega\epsilon_0 n^2 \beta} \frac{\partial^2 H_x}{\partial x \partial y} \\ E_y = -\frac{\omega\mu_0}{\beta} H_x - \frac{1}{\omega\epsilon_0 n^2 \beta} \frac{\partial^2 H_x}{\partial y^2} \\ E_z = \frac{j}{\omega\epsilon_0 n^2} \frac{\partial H_x}{\partial y} \\ H_z = \frac{-j}{\beta} \frac{\partial H_x}{\partial x} \end{cases} \quad (2.39)$$

The modes in Equations 2.36 and 2.37 are described as E_{pq}^x (p and q are integers starting from 1). And the modes in Equations 2.38 and 2.39 are described as E_{pq}^y . In both cases we obtain the same relations between the transverse wavenumbers k_x , k_y , γ_x , and γ_y .

$$\gamma_x^2 = k_0^2(n_1^2 - n_0^2) - k_x^2 \quad (2.40)$$

$$\gamma_y^2 = k_0^2(n_1^2 - n_0^2) - k_y^2 \quad (2.41)$$

The propagation constant is obtained from

$$\beta^2 = k_0^2 n_1^2 - (k_x^2 + k_y^2) \quad (2.42)$$

If we apply the boundary conditions that the electric field E_z is continuous at $x = w$, and H_z in $y = d$ we contain the following dispersion equation:

$$k_x w = (p - 1) \frac{\pi}{2} + \tan^{-1} \left(\frac{n_1^2 \gamma_x}{n_0^2 k_x} \right) \quad (2.43)$$

$$k_y d = (q - 1) \frac{\pi}{2} + \tan^{-1} \left(\frac{\gamma_y}{k_y} \right) \quad (2.44)$$

If we apply the boundary conditions that the magnetic field H_z is continuous at $x = w$, and E_z in $y = d$ we contain the following dispersion equation:

$$k_x w = (q - 1) \frac{\pi}{2} + \tan^{-1} \left(\frac{\gamma_x}{k_x} \right) \quad (2.45)$$

$$k_y d = (p - 1) \frac{\pi}{2} + \tan^{-1} \left(\frac{n_1^2 \gamma_y}{n_0^2 k_y} \right) \quad (2.46)$$

Chapter 3

Coupled Mode Theory

The Coupled Mode Theory (CMT) [18] [19] was introduced in 1950's for microwave devices, it was in the 1970's when its application extended to optical devices, specifically in dielectric waveguides. The theory is of great use in analyzing devices and predicting fundamental characteristics by simple analytic means, and tractable to computational devices. It is essential in the construction of practical optical devices dealing with the mutual light-wave interaction between two propagation modes. In the following chapter we will present the derivation of the coupled mode equations, and we will explain the details of important devices.

3.1 Coupled Mode Equations

In this section we will present the EM derivations of the coupled-mode equations [19] parting from the configuration of a basic slab dielectric waveguide. The emntioned waveguide consists of a film of thickness t and index of refraction n_2 sandwiched between 2 mediums, the cover with index n_1 and the substrate with n_3 .

3.1.1 TE Modes

Taking $\frac{\partial}{\partial y} = 0$, this waveguide can support a finite number of confined TE modes with field components E_y , H_x and H_z . The modes not confined in the inner layer are not considered. The field component E_y of the TE modes obeys the next wave equation:

$$\nabla^2 E_y = \frac{n_i^2}{c^2} \frac{\partial^2 E_y}{\partial t^2} \quad (3.1)$$

3.1. Coupled Mode Equations

Where $i = 1, 2, 3$ representing the three regions of the waveguide. We take $E_y(x, z, t)$ in the form:

$$E_y(x, z, t) = \mathcal{E}_y(x) \exp^{i(\omega t - \beta z)} \quad (3.2)$$

In which the transverse function $\mathcal{E}_y(x)$ is equal to:

$$\mathcal{E}_y(x) = \begin{cases} C \exp^{-(qx)}, & 0 \leq x < \infty \\ C[\cos(hx) - \frac{q}{h} \sin(hx)], & -t \leq x \leq 0 \\ C[\cos(ht) + \frac{q}{h} \sin(ht)] \exp^{p(x+t)}, & -\infty < x \leq -t \end{cases} \quad (3.3)$$

Which applied to each region we obtain:

$$\begin{aligned} h &= (n_2^2 k^2 - \beta^2)^{1/2} \\ q &= (\beta^2 - n_1^2 k^2)^{1/2} \\ p &= (\beta^2 - n_3^2 k^2)^{1/2} \\ k &= \frac{\omega}{c} \end{aligned} \quad (3.4)$$

The condition of continuity of components E_y and H_z is that it must be continuous in $x = 0$ and in $x = -t$ and so we get the next equation:

$$\tan(ht) = \frac{q + p}{h \left(1 - \frac{pq}{h^2}\right)} \quad (3.5)$$

Which is used with Equation 3.4 to determine the eigenvalues β of the confined TE modes.

If the mode with $E_y = A\mathcal{E}_y(x)$ which corresponds to a power flow of $|A|^2 W/m$ wants to be normalized we determine C with the next condition:

$$-\frac{1}{2} \int_{-\infty}^{\infty} E_y H_x^* dx = \frac{\beta_m}{2\omega\mu} \int_{-\infty}^{\infty} [\mathcal{E}_y^{(m)}(x)]^2 dx = 1 \quad (3.6)$$

$$C_m = 2h_m \left[\frac{\omega\mu}{|B_m| \left(t + \frac{1}{q_m} + \frac{1}{p_m}\right) (h_m^2 + q_m^2)} \right]^{\frac{1}{2}} \quad (3.7)$$

Since the $\mathcal{E}_y$ are orthogonal we obtain:

$$\int_{-\infty}^{\infty} \mathcal{E}_y^{(l)} \mathcal{E}_y^{(m)} dx = \frac{2\omega\mu}{\beta_m} \delta_{l,m} \quad (3.8)$$

3.1.2 TM Modes

Taking $\frac{\partial}{\partial y} = 0$, this waveguide can support a finite number of confined TM modes with field components H_y , E_x and E_z . The modes not confined in the inner layer are not considered. The field components of the TM modes are:

$$\begin{aligned} H_y(x, z, t) &= \mathcal{H}_y(x) \exp^{i(\omega t - i\beta z)} \\ E_x(x, z, t) &= \frac{i}{\omega\epsilon} \frac{\partial H_y}{\partial z} = \frac{\beta}{\omega\epsilon} \mathcal{H}_y(x) \exp^{i(\omega t - \beta z)} \\ E_z(x, z, t) &= -\frac{i}{\omega\epsilon} \frac{\partial H_y}{\partial x} \end{aligned} \quad (3.9)$$

The transverse function $\mathcal{H}_y(x)$ is:

$$\mathcal{H}_y(x) = \begin{cases} -C \left[\frac{h}{q} \cos(ht) + \sin(ht) \right] \exp^{p(x+t)}, & x < -t \\ C \left[-\frac{h}{q} \cos(hx) + \sin(hx) \right], & -t < x < 0 \\ -\frac{h}{q} C \exp^{-qx}, & x > 0 \end{cases} \quad (3.10)$$

The continuity of H_y and E_z requires that the propagation constants obey the eigenvalue equation:

$$\tan(ht) = \frac{h(\bar{p} + \bar{q})}{h^2 - \bar{p}\bar{q}} \quad (3.11)$$

Where

$$\bar{p} = \frac{n_2^2}{n_3^2} p \quad (3.12)$$

$$\bar{q} = \frac{n_2^2}{n_1^2} q$$

To normalize the constant so that the field carries 1 W per unit width in the y direction we must follow the next condition:

$$\frac{1}{2} \int_{-\infty}^{\infty} H_y E_x^* dx = \frac{\beta}{2\omega} \int_{-\infty}^{\infty} \frac{\mathcal{H}_y^2(x)}{\epsilon} dx = 1 \quad (3.13)$$

3.1. Coupled Mode Equations

where $n_1^2 = \epsilon_1/\epsilon_0$ so we obtain:

$$C_m = 2\sqrt{\frac{\omega\epsilon_0}{\beta_m t_{eff}}} \quad (3.14)$$

$$t_{eff} = \frac{\bar{q}^2 + h^2}{\bar{q}^2} \left[\frac{1}{n_2^2} + \frac{q^2 + h^2}{\bar{q}^2 + h^2} \frac{1}{n_1^2 q} + \frac{p^2 + h^2}{\bar{p}^2 + h^2} \frac{1}{n_3^2 p} \right]$$

3.1.3 The Coupling Equation

The wave equation obeyed by the unperturbed modes is:

$$\nabla^2 \mathbf{E}(\mathbf{r}, t) = \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (3.15)$$

We can represent the perturbation as a distributed polarization source $\mathbf{P}_{pert}(\mathbf{r}, t)$, which accounts for the deviation of the medium polarization from that which accompanies the unperturbed mode. The wave equation for the perturbed case follows directly from Maxwell's equations if we take $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$.

$$\nabla^2 E_y(\mathbf{r}, t) = \mu\epsilon \frac{\partial^2 E_y}{\partial t^2} + \mu \frac{\partial^2}{\partial t^2} [P_{pert}(\mathbf{r}, t)]_y \quad (3.16)$$

Similar equations correspond to the remaining components in the other directions.

If we take the eigenmodes of Equation 3.15 as an orthonormal set, and we assume slow variation so that $d^2 A_m/dz^2 \ll \beta_m dA_m/dz$, and recalling that $\mathcal{E}_y^{(m)}(x) \exp^{i(\omega t - \beta_m z)}$ obeys the unperturbed Equation 3.15 we obtain:

$$\sum_l \left[-i\beta_l \frac{dA_l}{dz} \mathbf{E}_y^{(l)}(x) \exp^{i(\omega t - \beta_l z)} \right] + c.c. = \mu \frac{\partial^2}{\partial t^2} (P_{pert})_y \quad (3.17)$$

Where l extends over the discrete set of confined modes and includes both positive and negative traveling waves.

Multiplying Equation 3.17 with $\mathbf{E}_y^{(m)}(x)$, integrating and making the orthogonality relation mention before we finally attain the following equation:

$$\frac{dA_m^{(-)}}{dz} \exp^{i(\omega t + \beta_m z)} - \frac{dA_m^{(+)}}{dz} \exp^{i(\omega t - \beta_m z)} + c.c. = \frac{-i}{2\omega} \frac{\partial^2}{\partial t^2} \int_{-\infty}^{\infty} [P_{pert}(\mathbf{r}, t)]_y \mathcal{E}_y^{(m)}(x) dx \quad (3.18)$$

Where $A_m^{(-)}$ is the complex mode amplitude of the negative traveling TE mode while $A_m^{(+)}$ is the positive traveling TE mode.

Chapter 4

MMI Self-Imaging Model

The operation of any optical MMI device is based on the self-imaging principle [4] of periodic objects illuminated by coherent light, this principle was first described as a property of multimode waveguides by which an input field profile is reproduced in single or multiple images at periodic intervals along the propagation direction of the guide.

4.1 MultiMode Waveguides

The main structure of the MMI device is a multimode waveguide [4], which has the property of being able to propagate more than 3 modes. To excite these modes it is necessary to incorporate to the mentioned structure the input and output segments or waveguides which in will be singlemode.

For an analytic point of view a full Modal Propagation Analysis (MPA) is of great use since the field distribution of the waveguide modes can be determined. To be allowed to easily optimize the device structure it is of essential use the BPM method.

4.1.1 Propagation Constants

The multimode waveguide is composed by a ridge of an effective refractive index of n_r of width W_M and a cladding with a effective refractive index of n_c as shown in Figure 4.1 .

The waveguide is able to propagate m lateral modes with modes numbers $v = 0, 1, \dots (m-1)$ as shown in Figure 4.2

The propagation constant β_v and the lateral wavenumber $k_y v$ are related to the ridge index by the dispersion equation:

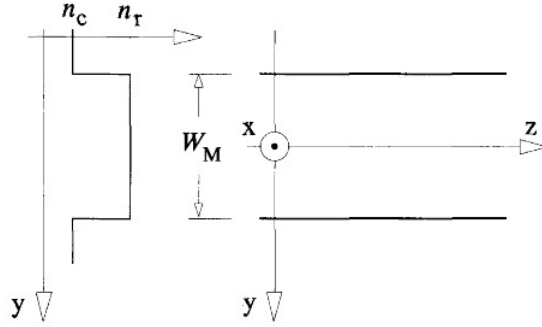


Figure 4.1: 2D representation of the refractive step index profile and top view of the multimode waveguide [4]

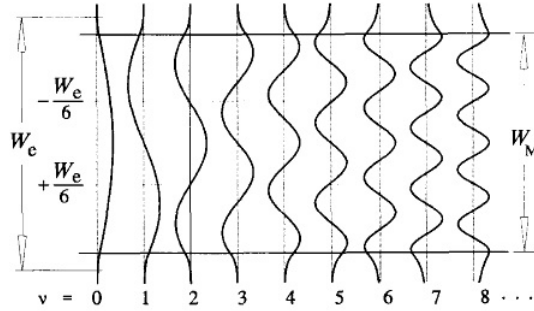


Figure 4.2: Amplitude-normalized lateral field profiles [4]

$$k_y v^2 + \beta_v^2 = k_0^2 n_r^2 \quad (4.1)$$

where

$$\begin{aligned} k_0 &= \frac{2\pi}{\lambda_0} \\ k_y v &= \frac{(v+1)\pi}{W_e v} \end{aligned} \quad (4.2)$$

Where the effective width $W_e v$ takes into account the polarization dependent lateral penetration depth of each mode field, associated with the Goos-Hähnchen shift at the ridge boundaries. In general the effective widths $W_e v$ can be approximated by the effective width $W_e 0$ corresponding to the fundamental mode.

$$W_e v \simeq W_e = W_M + \left(\frac{\lambda_0}{\pi} \right) \left(\frac{n_c}{n_r} \right)^{2\sigma} (n_r^2 - n_c^2)^{-1/2} \quad (4.3)$$

Where $\sigma = 0$ for TE and $\sigma = 1$ for TM polarization. By using binomial expansion with $k_y v^2 \ll k_0^2 n_r^2$, the propagation constant can be deduced into:

$$\beta_v \simeq k_0 n_r - \frac{(v+1)^2 \pi \lambda_0}{4 n_r W_e^2} \quad (4.4)$$

Therefore, the propagation constants in a step-index multimode waveguide show a nearly quadratic dependence with respect to the mode number v . By defining L_π as the beat length of the two lowest order modes:

$$L_\pi = \frac{\pi}{\beta_0 - \beta_1} \simeq \frac{4 n_r W_e^2}{3 \lambda_0} \quad (4.5)$$

And the propagation constants spacing can be expressed as:

$$(\beta_0 - \beta_1) \simeq \frac{v(v+2)\pi}{3L_\pi} \quad (4.6)$$

4.1.2 Guided Mode Propagation Analysis

An input profile $\Psi(y, 0)$ imposed at $z = 0$ and totally contained within W_e will be decomposed into the modal field distributions $\psi_v(y)$ of all modes as shown in Figure 4.3 .

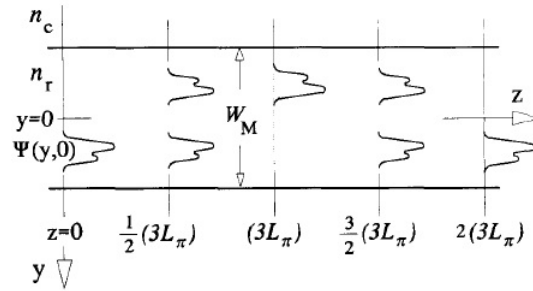


Figure 4.3: Input field and mirrored images in the multimode waveguide [4]

$$\Psi(y, 0) = \sum_v c_v \psi_v(y) \quad (4.7)$$

Where the summation should be understood as including guided as well as radiative modes. The field excitation coefficients c_v can be estimated by using overlap integrals:

$$c_v = \frac{\int \Psi(y, 0) \psi_v(y) dy}{\sqrt{\int \psi_v^2(y) dy}} \quad (4.8)$$

If the spatial spectrum of the input field is narrow enough to avoid the excitation of unguided modes, it can be decomposed into guided modes alone.

$$\Psi(y, 0) = \sum_{v=0}^{m-1} c_v \psi_v(y) \quad (4.9)$$

From which we can determine the field at a distance z that can be written as a superposition of all the guided mode field distributions:

$$\Psi(y, z) = \sum_{v=0}^{m-1} c_v \psi_v(y) \exp^{j(\omega t - \beta_v z)} \quad (4.10)$$

Or

$$\Psi(y, z) = \sum_{v=0}^{m-1} c_v \psi_v(y) \exp^{j(\beta_0 - \beta_1)z} \quad (4.11)$$

Therefore, substituting Equation 4.6 we obtain the expression with distance $z = L$.

$$\Psi(y, L) = \sum_{v=0}^{m-1} c_v \psi_v(y) \exp \left[j \left(\frac{v(v+2)\pi}{3L_\pi} \right) L \right] \quad (4.12)$$

The different images formed will be determined by the modal excitation c_v , and the properties of the mode phase factor.

$$\exp \left[j \frac{v(v+2)\pi}{3L_\pi} L \right] \quad (4.13)$$

4.2 General Interference

Self-Imaging interference mechanism in which the images are independent of the modal excitation, in this category we have the formation of both the single and the multiple images.

4.2.1 Single Images

The single images are those that are a replica of the input field, meaning the single image obtained at $\Psi(y, L)$ will be equal to the image of $\Psi(y, 0)$, if:

$$\exp \left[j \frac{v(v+2)\pi}{3L_\pi} L \right] = 1 \quad \text{or} \quad (-1)^v \quad (4.14)$$

The first condition refers to the phase changes of all the modes along L and that it must differ by integer multiples of 2π . The replica of the input occurs due to the fact that all the guided modes interfere with the same relative phases, these images are called also direct image.

The second condition means that the phase changes must alternate between odd and even multiples of π . This produces the existence of even modes that will be in phase with the input and the odd modes which will be the anti-phase of the same.

Due to the odd symmetry the interference produces a mirrored image with respect to the $y = 0$ plane. We can see this characteristic in the following Equation:

$$\psi_v(-y) = \begin{cases} \psi_v(y) & \text{for } v \text{ even} \\ -\psi_v(y) & \text{for } v \text{ odd} \end{cases} \quad (4.15)$$

and in

$$v(v+2) = \begin{cases} \text{even for } v \text{ even} \\ \text{odd for } v \text{ odd} \end{cases} \quad (4.16)$$

Following the property shown in Equation 4.16 and taking into account the conditions in Equation 4.14 we can deduce that it will be fulfilled at:

$$L = p(3L_\pi) \quad \text{with } p = 0, 1, 2, \dots \quad (4.17)$$

Where p denotes the periodic nature of the imaging through the multimode waveguide. Therefore, the direct and mirrored single images of the input field $\Psi_v(y, 0)$ will be formed by general interference at a z distance which are, respectively, even and odd multiples of the length $(3L_\pi)$.

4.2.2 Multiple Images

Multiple images are also obtained as a result of the input field in $\Psi_v(y, 0)$, these are obtained between the single images determined to be fulfilled at distances given by Equation 4.17 .

If we consider uniquely the images positioned half-way between the direct and the mirrored images, meaning the following positions:

$$L = \frac{p}{2}(3L_\pi) \quad \text{with } p = 1, 3, 5, \dots \quad (4.18)$$

In which we can find the fields by substituting the length into the field profile,

$$\Psi(y, \frac{p}{2}3L_\pi) = \sum_{v=0}^{m-1} c_v \psi_v(y) \exp \left[jv((v+2)p(\frac{\pi}{2})) \right] \quad (4.19)$$

Where p is an odd integer, and where if applied the conditions and properties of symmetry we obtain:

$$\begin{aligned} \Psi(y, \frac{p}{2}3L_\pi) &= \sum_{v \text{ even}} c_v \psi_v(y) + \sum_{v \text{ odd}} (-j)^p c_v \psi_v(y) \\ &= \frac{1 + (-j)^p}{2} \Psi(y, 0) + \frac{1 - (-j)^p}{2} \Psi(-y, 0) \end{aligned} \quad (4.20)$$

These pair of images obtained from the input field are in quadrature and have an amplitude of $1/\sqrt{2}$, also they are separated between each other by a distance in the propagation direction of $z = \frac{1}{2}(3L_\pi), \frac{3}{2}(3L_\pi), \dots$. This is essential and can be used to produce the 2x2 3-dB coupler.

As seen, the multi-fold images are formed at intermediate z positions. The positions and the phases of the N-fold images at a certain z distance is calculated by using Fourier analysis and properties of generalized Gaussian sums. To perform this analysis we introduce a field $\Psi_{in}(y)$ as an extension of the input field, asymmetric respect to the $y = 0$ plane and with periodicity of $2W_e$.

$$\Psi_{in}(y) = \sum_{v=-\infty}^{\infty} [\Psi(y - v2W_e, 0) - \Psi(-y + v2W_e, 0)] \quad (4.21)$$

We can approximate the mode field amplitudes to a sine-like function

$$\psi_v(y) \simeq \sin(k_{yv}y) \quad (4.22)$$

Permitting to considerate it as an Fourier expansion at distances

$$L = \frac{p}{N}(3L_\pi) \quad (4.23)$$

Where $p \geq 0$ and $N \geq 1$ are integers with no common divisors, and the field will be expressed as follows:

$$\Psi(y, L) = \frac{1}{C} \sum_{q=0}^{N-1} \Psi_{in}(y - y_q) \exp(j\varphi_q) \quad (4.24)$$

where

$$y_q = p(2q - N) \frac{W_e}{N} \quad (4.25)$$

$$\varphi_q = p(N - q) \frac{q\pi}{N} \quad (4.26)$$

And C is a complex normalized constant with $|C| = \sqrt{N}$, p refers to the imaging periodicity along the direction of z , and q represents each of the N images along the y direction.

The mentioned equations show that, at distances $z = L$, N images are formed of the extended field $\Psi_{in}(y)$ located at the position y_q , each with amplitude $1/\sqrt{N}$ and phase φ_q . This leads to N images, generally not equally spaced between them, of the input field, being formed inside the physical guide and within the lateral boundaries.

The multiple self-imaging mechanism allows the realization of NxN or NxM optical couplers. Shorter devices are obtained with $p = 1$. Optical phases for the NxN case is given by the following:

$$\varphi_{rs} = \frac{\pi}{4N}(s - 1)(2N + r - s) + \pi \quad \text{for } r + s \text{ even} \quad (4.27)$$

$$\varphi_{rs} = \frac{\pi}{4N}(r + s - 1)(2N - r - s + 1) + \pi \quad \text{for } r + s \text{ odd} \quad (4.28)$$

where $r = 1, 2, \dots, N$ is the numbering of the input waveguides (bottom-up) and $s = 1, 2, \dots, N$ is the numbering of the output waveguides (top-down).

4.3 Restricted Interference

MMI Couplers permit the restriction of the modal excitation [4], meaning, it is possible to excite, by the input fields, only some of the guided modes in the multimode waveguide. This selective excitation reveals interesting multiplicities of $v(v + 2)$ which allow new interference mechanisms through shorter periodicities of the mode phase factor.

4.3.1 Paired Interference

The selective excitation of modes can offer us certain advantages, like for example when:

$$\text{mod}_3[v(v + 2)] = 0 \quad \text{for } v \neq 2, 5, 8, \dots \quad (4.29)$$

We reduce the length periodicity of the mode phase factor 3 times the original size, but also the following condition must be taken into account.

$$c_v = 0 \quad \text{for } v = 2, 5, 8 \quad (4.30)$$

Resulting in lengths of direct and inverted images, of the input field, determined by the next equation:

$$L = p(L_\pi) \quad \text{with } p = 0, 1, 2, \dots \quad (4.31)$$

As long as the mentioned modes are not excited in the multimode waveguide. Parting from this point we can determine the length of N-fold images:

$$L = \frac{p}{N}(L_\pi) \quad (4.32)$$

where $p \geq 0$ and $N \geq 1$ are integer with no common divisors.

Now that we have explained the advantage, the procedure to excite the wished modes is detailed. To obtain this behavior we must launch an even symmetric input field $\Psi(y, 0)$, usually a Gaussian beam, at $y = \pm W_e/6$. Position where the modes $v = 2, 5, 8, \dots$ present a zero with odd symmetry. The overlap of the integrals of the field excitation coefficient c_v between the symmetric input and the asymmetric mode fields will vanish and therefore we will obtain $c_v = 0$ for $v = 2, 5, 8, \dots$, in this case the input waveguides are limited to 2.

When the selective excitation is performed, the modes contributing to the imaging will be paired, the mode pairs 0-1, 3-4, 6-7 and so on, which will have similar properties, based on this fact they obtain the name paired interference.

A very used case of this restricted interference is the 2x2 MMI Coupler or the Two Mode Interference (TMI), commonly used in silica based dielectric rib type waveguides.

4.3.2 Symmetric Interference

This selective excitation of a 1-to-N beam splitter in which the modes excited are the even symmetrical can produce the shortage of four times the original result of the waveguides length. As in the previous restricted interference we will proceed to explain through an example, where:

$$\text{mod}_4[v(v+2)] = 0 \quad \text{for } v \text{ even} \quad (4.33)$$

Having a reduction of the length periodicity of the mode phase when the following condition is fulfilled:

$$c_v = 0 \quad \text{for } v = 1, 3, 5, \dots \quad (4.34)$$

Consequently the direct and inverted single images of the input field $\Psi(y, 0)$ will be obtained at the length of:

$$L = \frac{p}{N} \left(\frac{3L_\pi}{4} \right) \quad (4.35)$$

With N images of the input field, symmetrically located along the y -axis with equal spacing W_e/N .

In the symmetrically excited MMI Couplers at half the self imaging length we can observe the formation of the 2-fold image, and as the length diminishes the number of images increases accordingly to the Equation 4.35 , until they are no longer resolvable.

We can generalize that to obtain a low-loss well-balanced 1-to- N splitting of a Gaussian beam the multimode waveguide must support at least $m = N + 1$ modes. The most common and simple of symmetric interference is the 1x2 MMI coupler, needing only 2 symmetric modes.

Chapter 5

Fabrication Techniques

Throughout this chapter the process of fabrication of a Silicon Waveguide Devices [5] will be explained in detail, passing through the diversity of choices in methods to obtain the characteristics of the mentioned devices. The usage of this material in the fabrication of photonic devices is due to the low primary cost of the material, the mature and well characterized processing techniques that have been highly researched, developed and manufactured in the microelectronic industry which present an enormous advantage and permits the future fusion of both industries.

5.1 Silicon-on-Insulator (SOI)

SOI is defined as the most common structure of silicon photonic devices fabricated, it is composed by a thin layer of crystalline silicon on a insulating layer. From the mentioned me can underline the silicon-on-silicon dioxide (SiO_2), composed by a uniform layer of SiO_2 which is sandwiched between a thick silicon substrate and a thin surface layer of crystalline silicon (Figure 5.1).

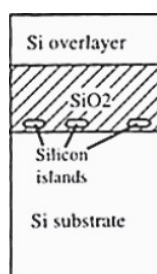


Figure 5.1: Schematic of Silicon-on-silicon dioxide [5]

5.1. Silicon-on-Insulator (SOI)

The SiO_2 has a refractive index of 1.46, and the crystalline silicon layer of 3.5, considerably higher and there for generating a specific type of SOI forming a classic waveguide form. Both layers of crystalline silicon and the buried silicon dioxide layer are characterized by having a thickness in the order of microns, but its value can change depending of the method of fabrication. The most commonly used are the explained in the following subsections.

5.1.1 Separation by IMplanted OXygen (SIMOX)

The SIMOX [5] consists on the implantation of a large amount of oxygen ions below the surface of a silicon wafer (Figure 5.2). Its simplicity is a factor of relevance that qualifies it as the most common method for the fabrication of large volumes of SOI material.

To describe the total amount of any ion species implanted into a wafer the implanted ion dose is used. The ion dose is the total number of ion that pass through one square centimeter of the wafer surface, this is measured in units of ions/cm^2 .

In the SIMOX process it is required to present a total implantation dose of over 10^{18}cm^{-2} , and under normal temperatures we will obtain unwanted amorphous silicon over-layers, therefore it must be kept at a temperature of 600°C during the implantation on the silicon substrate.

In this process we obtain a certain depth of the SiO_2 , and hence the thickness of the silicon overlayer, through the energy used to implant the oxygen ions into the crystalline silicon. The mentioned energy can go up to 200keV , and the consequences in depth from oxygen variation are viewed in the profiles showed in Figure 5.3 .

Where we can observe certain behaviors, in the case where we have the application of less than 10^{16}cm^{-2} we get as a profile a gaussian function. As the oxygen dose increases, the peak concentration of ions (O^+) saturates to a concentration of that found in stoichiometric SiO_2 , with further implantation over 10^{18}cm^{-2} , the oxygen profile begins to flatten forming a buried and continuous layer of SiO_2 . and finally the silicon wafer is annealed at a temperature of 1300°C for several hours, this produces a uniform buried SiO_2 layer with distinct interfaces with 2 adjacent silicon layers. The annealing insures the silicon overlayer is denuded of implantation related, primary lattice defects.

The concentration of secondary defects in the silicon overlayer is of great importance to silicon photonics, and the micro-roughness of the silicon overlayer surface and the overlayer/buried oxide interface.

5.1.2 Bond and Etch-back SOI (BESOI)

This process is the result of the use of a phenomenon produced by the intimate contact of two hydrophilic surfaces [5], such as SiO_2 , that creates a highly strong bond between them. The process follows three steps which are shown in Figure 5.4 .

The steps are following:

- (a) Oxidation of 2 wafers to be bonded.

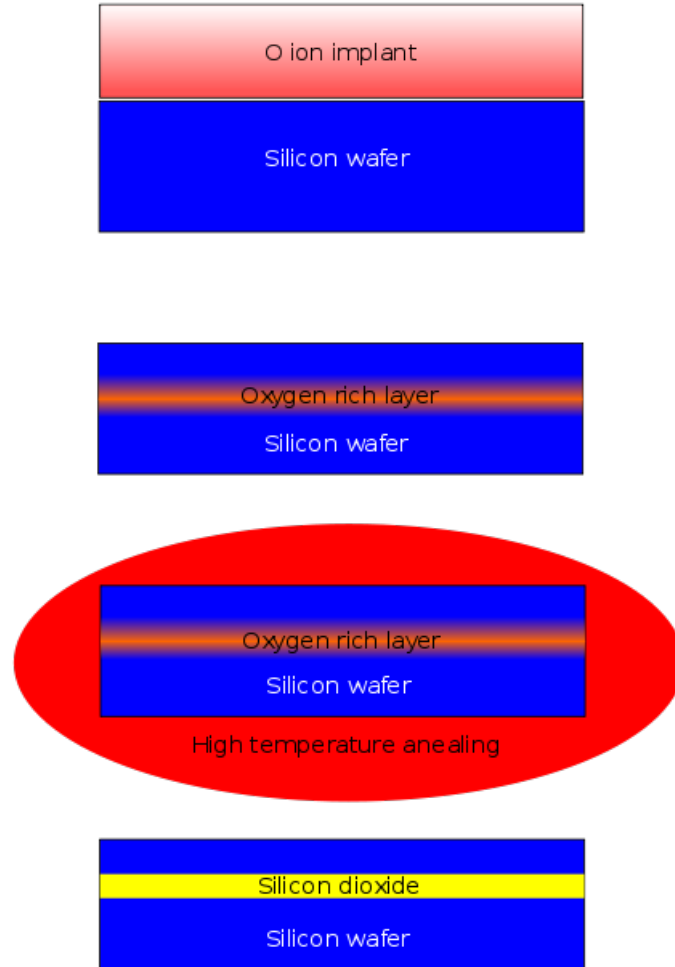


Figure 5.2: SIMOX processing schematic [6]

- (b) Formation of the chemical bond.
- (c) Thinning (etching) of one of the wafers.

The bonding process is done by bringing the wafers into contact at room temperature, where the initial bond is formed. The bond strength is increased to that of bulk material via subsequent thermal processing to temperatures as high as 1100 °C.

The etching or wafer thinning uses the Chemical Mechanical Polishing (CMP) technique, commonly used in microelectronics for wafer planarization. CMP requires that the wafer surface be both weakened and subsequently removed during a single processing step.

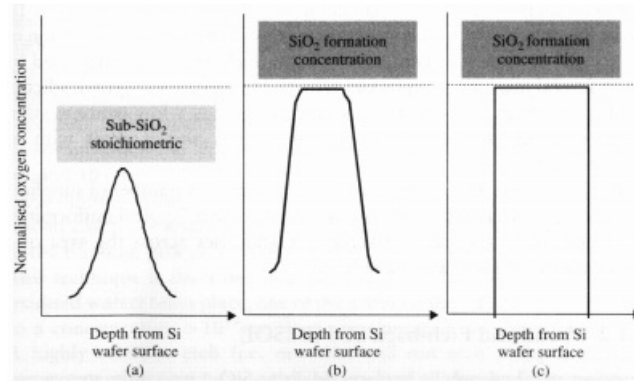


Figure 5.3: Variation of oxygen profile during the SIMOX process [5]

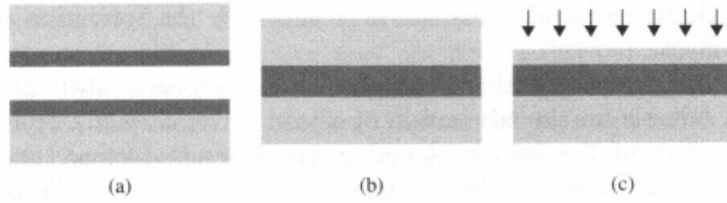


Figure 5.4: BESOI process [5]

- The silicon surface to be polished is brought into contact with a rotating pad, and simultaneously a chemically reactive slurry containing an abrasive component, such as alumina and glycerin, weakens and removes surface layers.
- The process removes the majority of the polished bonded wafer, leaving a thin silicon overlayer on a buried SiO_2 layer, supported by a silicon substrate.
- The removing method limits the achievable thickness for the silicon overlayer to around 10 microns.

An improvement in SOI thickness uniformity can be achieved by the use of an End-Stop in the thinning process reducing or even eliminating the need for CMP. This improvement is based on the application of the following modification.

1. Subsequently from the creation of the heavily doped p-type layer, a further, undoped intrinsic layer is epitaxially grown on the wafer surface.
2. Afterwards, a second non-selective etched process is used to remove the exposed p-type layer following the selective etch.
3. The final wafer is then a structure of one undoped silicon overlayer on the buried SiO_2 .

5.1.3 Wafer Splitting (SmartCut Process to produce Unibond Wafers)

It is considered to be the fusion of the SIMOX and the BESOI [6] [5]. The steps that describe the process are shown in Figure 5.5 .

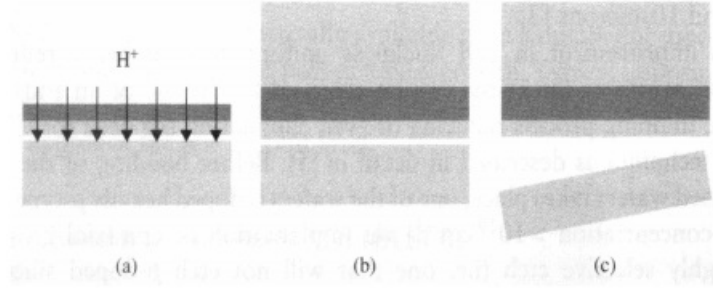


Figure 5.5: Smart Cut process [5]

- (a) Thermally oxidized wafer is implanted with a high dose of hydrogen, approximately 10^{17}cm^{-2} . The implanted hydrogen ions form a gaussian-like profile. The distance from the wafer surface of the peak of the profile depends on the H^+ ion energy, usually between a few hundred nanometers and a few microns. The hydrogen ions, and the silicon lattice damage caused by the stopping of the ions, are at their greatest concentration at this depth, and here the silicon lattice bonds are significantly weakened.
- (b) A second wafer (which may or may not have a thermal SiO_2 -covered surface) is bonded to the first as in the BESOI process.
- (c) The thermal processing at 600°C and 1100°C splits the implanted wafer at a point consistent with the range of hydrogen ions. A fine CMP is employed (Figure 5.6) to reduce roughness at the SOI surface.

To increase the thickness of the silicon overlayer can be obtained by the use of the epitaxial silicon growth. The non-uniformity of the position of the implanted hydrogen profile peak and therefore the overlayer thickness are only a few percent. Although the overlayer receives a high dose implant, the small mass of the hydrogen ion ensures that negligible residual damage remains at the end of the thermal processing.

5.1.4 Silicon Epitaxial Growth

Epitaxial means that the grown layer is a ordered mono-crystal [5], essential if efficient opto-electrical devices are to be fabricated.

The use of epitaxial silicon as the waveguiding medium has the additional advantage of doping and defect levels below those found in wafers cut from an ingot following bulk growth using the Czochalski (Cz) or Floating Zone (FZ) methods.

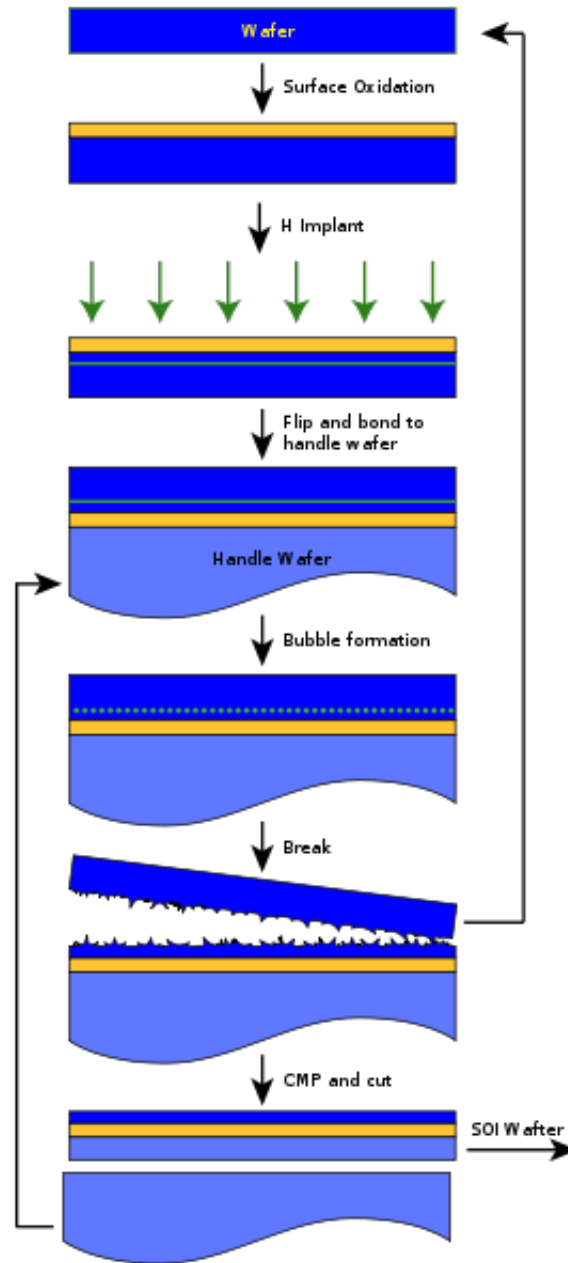


Figure 5.6: Smart Cut detailed process (sub-steps)[6]

The most common epitaxial silicon growth technique is Chemical Vapor Deposition (CVD), process which deposits a solid film on the surface of a silicon wafer by the reaction of a gas mixture at that surface. For silicon deposition, dichlorosilane (SiH_2Cl_2) is often used as the source gas.

The wafer surface must be raised in temperature (over 1000°C) to create the chemical reaction. The desired thickness of the film required in silicon waveguide fabrication dictates the use of vapor phase epitaxy, although a solid source can also be used with Molecular Beam Epi-

taxy (MBE). The resulting non-uniformity by this process is less than 1%.

5.2 Fabrication of Surface Etched Features

To this point the fabrication of SOI material, structures that are used in the guidance of the light, but in this section we will approach features that provide lateral confinement to the mentioned structures, characteristic of great importance will designing the slab the waveguide. The fabrication steps in forming the rib and other guiding structures in the silicon overlayer will be explained in this section.

5.2.1 Photolithography

The control of the photo process is one of the most important factors in silicon photonic fabrication since the width of the rib waveguide is determined by its photolithographical characteristics.

To control the dimensions and obtain a minimal feature size at a level of $10nm$ the Critical Dimension (CD) is used, this control is in excess of that required to form the most basic of silicon photonic structures such as large-cross-section and the singlemode silicon rib.

5.2.1.1 Wafer Preparation

This process is basically the elimination of contaminate particles on the surface of the wafer, and later on it must be desorbed of any moisture. The latest is a great necessity since the cleaning process is done via a wet process, ending in a DI wafer rinse and dry, to achieve this the wafer is baked at $150\text{ }^{\circ}\text{C}$, then the wafer is coated with an adhesion promoter (hexamethyldisilazane).

5.2.1.2 Photoresist Application

To create a better control when fabricating exotic devices or waveguides with dimensions which are submicron photolithography transfers a mask-defined pattern to the wafer by printing on its surface using a photosensitive polymer called photoresist.

The application is performed by first coating the wafer with liquid photoresist following preparation. The resist is dispensed on the center of the wafer which is held via a vacuum seal on a metal or polymer chuck. When approximately 10 ml of this liquid has been dispensed, the wafer is spun at a typical speed of 1-5 krmp, this distributes the resist over the entire surface of the wafer (Figure 5.7).

5.2.1.3 Soft Bake

A post-spin soft bake is used to drive off most of the solvents in the resist while at the same time improving resist uniformity and adhesion, this is performed at $100\text{ }^{\circ}\text{C}$ for a few minutes.

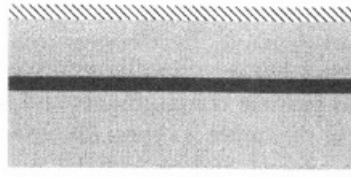


Figure 5.7: The SOI wafer is uniformly coated with a thin polymer known as photoresist [5]

5.2.1.4 Exposure to Ultraviolet Light

The wafer is transferred to the mask-aligner where it is placed, with sub-micron precision, relative to the permanent pattern defined on the mask. Unless this is the first wafer layer, the pattern will be integrated with all previous layers. Once correctly aligned, the wafer is exposed to UV light (Figure 5.8).

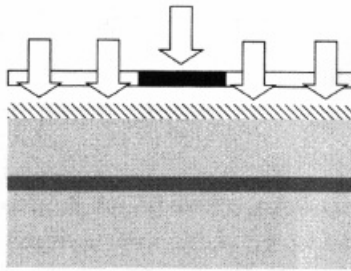


Figure 5.8: The resist is exposed to UV light through a permanent mask. The mask shown here is designed to result in waveguide formation [5]

When the process performed is *Positive Resist* the light passes through the transparent regions of the mask and activates the photosensitive components of the resist, such that these areas of resist are removed during the developing stage as shown in Figure 5.9 .

In the *Negative Resist* process the unexposed areas are removed.

5.2.1.5 Photoresist Developing

At this stage the pattern is created, where the wafer is exposed to a developing solution. Whether the process is positive or negative resist, the solution will dissolve the activated resist, or the un-activated, leaving the resist pattern intact.

5.2.1.6 Hard Bake

This process drives off the remaining resist solvents and further strengthens the resist adhesion to the wafer surface. It is typically performed at a temperature of 90-140 °C for several minutes,

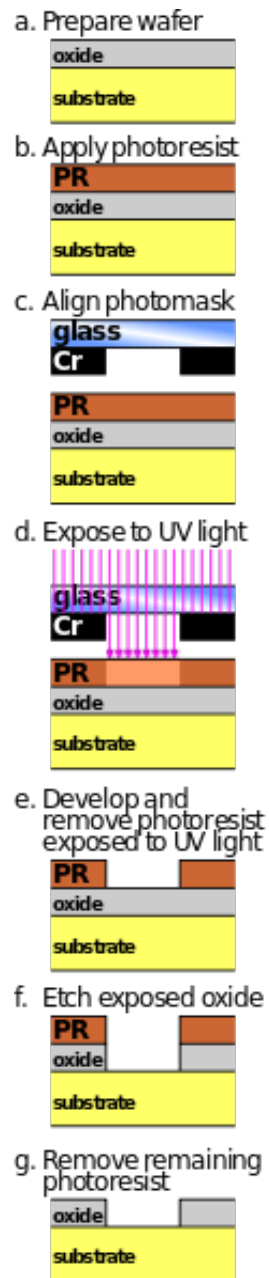


Figure 5.9: Dry etching using positive photoresist during a photolithography process [7]

the upper limit of the mentioned temperatures must be such that the hardbake does not result in pattern deformation via resist flow. Following hardbake, the desired pattern is printed in the photoresist (Figure 5.10).

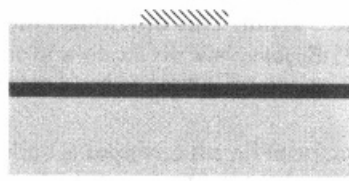


Figure 5.10: Following hardbake, the desired pattern is printed in the photoresist ready for transfer to the wafer [5]

5.2.2 Silicon Etching

Silicon Etching is divided in two categories, Wet Etching and Dry Etching, both have great advantages and disadvantages, but to reproduce features of submicron dimensions the Dry Etching approach is the best option. The wet etching employs a liquid chemical agent to remove the uppermost layer of the substrate in the areas that are not protected by the photoresist, the dry etching performs the same procedure but by the use of a plasma chemical agent instead.

When etching two characteristics [8] are of great importance, Selectivity (Figure 5.11) and Isotropy (Figure 5.12).

Selectivity If the etch is intended to make a cavity in a material, the depth of the cavity may be controlled approximately using the etching time and the known etch rate. More often, though, etching must entirely remove the top layer of a multilayer structure, without damaging the underlying or masking layers. The etching system's ability to do this depends on the ratio of etch rates in the two materials.

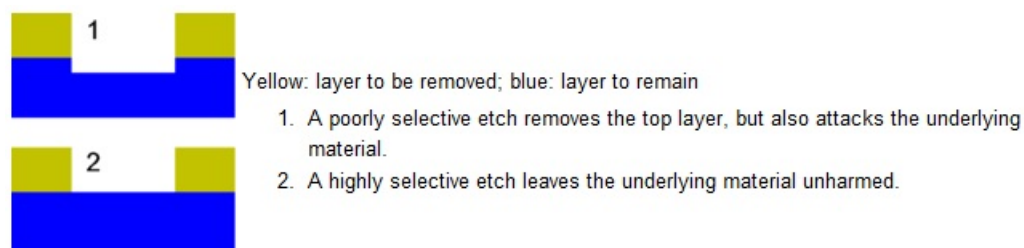


Figure 5.11: Selectivity [8]

Isotropy Some etches undercut the masking layer and form cavities with sloping sidewalls. The distance of undercutting is called bias. Etchants with large bias are called isotropic, because they erode the substrate equally in all directions. Modern processes greatly prefer anisotropic etches, because they produce sharp, well-controlled features.

Low-loss waveguides have been produced in both approaches, having small dimensions (typically $> 1\mu m$), and having characteristics like flexible process capability, tight tolerances and reproducible production.

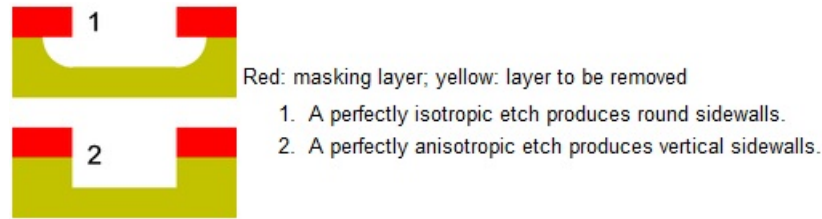


Figure 5.12: Isotropy [8]

Dry etching [8] proceeds through the formation of a low pressure plasma (ionized gas, virtually neutral, consisting of electrons, ions and mostly neutral particles). The process is favored since this technique can be made anisotropic, implying identical properties in all directions, in order to avoid significant undercutting of the photoresist pattern.

The formation of this localized plasma can be achieved by applying either Alternating Current (AC) or Direct Current (DC) power to a process gas contained in an isolated chamber. DC bias is rarely used in practice because it is incompatible with insulating electrodes causing surface charging and therefore an unstable plasma.

DC bias is rarely used in practice because it is incompatible with insulating electrodes causing surface charging and therefore an unstable plasma. The AC power on the other hand with charge build-up in one half-cycle, followed by charge neutralization in the next half will avoid this problem.

AC plasma generation (Figure 5.13) therefore dominates in the design of semiconductor processing equipment, with the most common frequency being 13.56 MHz.

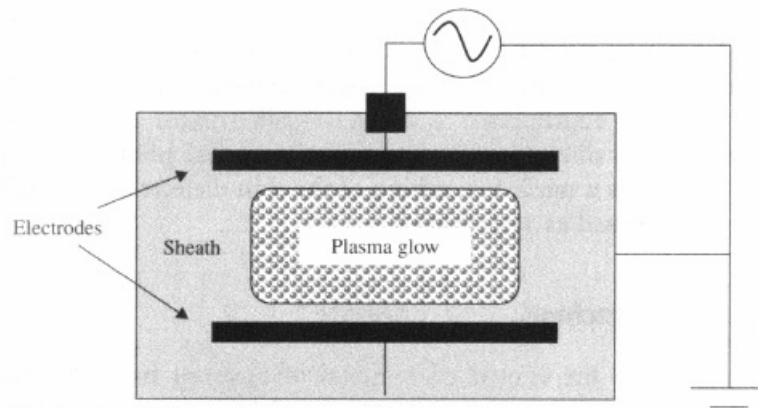


Figure 5.13: Schematic of a confined AC generated plasma [5]

The plasma gas (typically derived from CF_4) is introduced into the evacuated chamber, and stabilized at a pressure between 10^{-3} and 10^{-2} Torr. The initial AC signal is applied causing the high acceleration of the free electrons in the gas, which lose energy through interaction with the

1. The average value of a given critical dimension relative to the specialized value.
2. The uniformity of the critical dimension.
3. The linearity of feature sizes down to some lower limit.

The accuracy of the linewidth measuring tool and the cross-section profile of the features play an important role in determining the mentioned parameters, each of which has a unique effect on the wafer processing, as described ahead.

The correct average measure size (Parameter 1) is important to maintain the wafer exposure level that gives maximum exposure latitude. Biasing the wafer exposure to compensate for a large deviation in the average mask CD reduces the available exposure latitude. Also, large variations from one mask level to another requires re-calibrating the wafer exposure level. The CD uniformity (Parameter 2) is important to yield maximum circuit performance. The importance of feature size linearity (Parameter 3) is dependent on the circuit design.

All of the steps in forming a silicon rib will effect a change in the dimensions of the rib, and each requires characterization and constant monitoring. In high-volume manufacture this will entail the use of Statistical Process Control (SPC).

The dimensions critical to waveguide performace are described in Figure 5.15 .

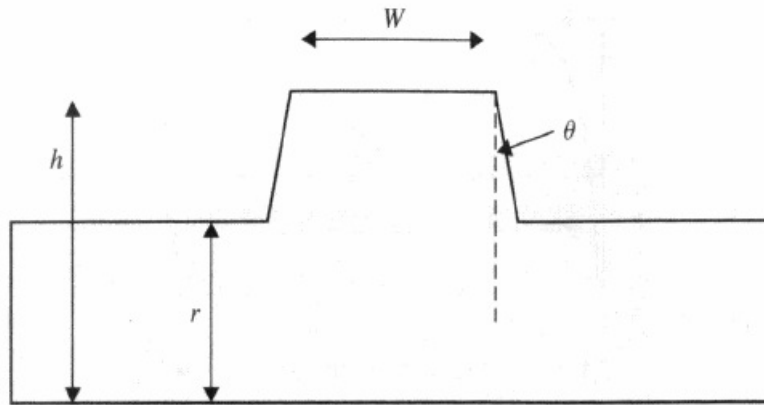


Figure 5.15: Schematic of a silicon waveguide. The dimensions critical to device performance are highlighted: rib width (W), silicon overlayer thickness (h), silicon thickness following rib etch (r) and rib wall angle (θ) [9]

The silicon overlayer thickness (h) is determined primarily by the process used to produce the SOI waveguiding layer. Silicon-on-SiO₂ lends itself particularly well to optical characterization, such as that performed with an infrared reflectometer or ellipsometer, and hence material may be screened easily prior to the fabrication of silicon photonic circuitry.

The width of the rib (W) and the rib wall angle (θ) depend on both the photolithography and the silicon etch. In the case of W , a process bias will exist at each process step. For

instance, the width of the rib image on the photo hard mask is generally different from that produced in the developed photoresist, and the ultimate width of the etched rib will be smaller than the photoresist image. The rib height ($h - r$) is determined by the dry silicon etch. Optical absorption and interferometric techniques exist allowing the determination of etch depth with a considerable degree of accuracy so-called end-point detection. Subsequent to etch a useful method for measuring an etch depth of micron dimensions is via surface profilometry.

Finally, an important consideration in calculating process bias for waveguides which have an upper oxide cladding is the reduction in h , r and W following thermal oxidation. Because thermal oxidation proceeds via a reaction of oxygen and silicon atoms, the effect is a consumption of silicon and hence reduction of critical dimensions. A comparison of the atomic density of silicon ($5 \times 10^{22} \text{cm}^{-3}$) and the molecular density of SiO_2 ($2.2 \times 10^{22} \text{cm}^{-3}$) leads immediately to the fact that, for the growth of a given thickness t of SiO_2 , a thickness of silicon equal to $(0.44t)$ is consumed.

5.3 MMI Device Fabrication and Innovative Fabrication Techniques

MMI devices, which are based on the structure of a multimode waveguide, are commonly obtained through the use of fabrication techniques with great precision such as the SIMOX method explained in Section 5.1.1, and surface etched features such as Photolithography and Dry Etching detailed in Sections 5.2.1 and 5.2.2. But the increasing scalability of these components have driven us to the use of more innovative techniques that will offer as result a high accurate structure at the nanometer level. In the following sections these techniques will be explained.

5.3.1 Electron-beam Lithography

Electron beam lithography [10], also called e-beam lithography, is the practice of emitting a beam of electrons (Figure 5.16) in a patterned fashion across a surface covered with a film called the resist, exposing the resist and through selectively removing either exposed or unexposed regions of the resist, this procedure is referred to as the developing stage. The purpose, as with photolithography, is to create very small structures in the resist that can subsequently be transferred to the substrate material, often by etching. Thanks to this feature we obtain the creation nanotechnology architectures (Figure 5.17).

The primary advantage of electron beam lithography is that it is one of the ways to beat the diffraction limit of light and make features in the nanometer regime. This form of maskless lithography has found wide usage in photomask-making used in photolithography, low-volume production of semiconductor components, and research and development.

The key limitation of electron beam lithography is throughput, the very long time it takes to expose an entire silicon wafer or glass substrate. A long exposure time leaves the user vulnerable to beam drift or instability which may occur during the exposure. Also, the turn-around time for reworking or re-design is lengthened unnecessarily if the pattern is not being changed the second time.

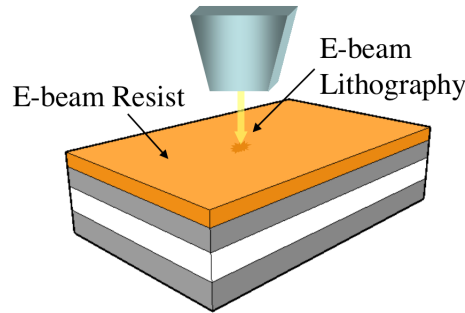


Figure 5.16: Electron Beam Lithography. Electron emission [10]

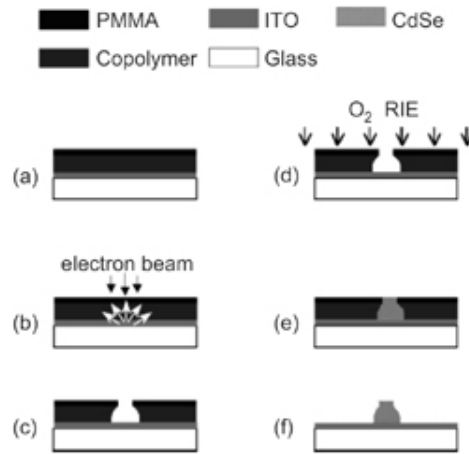


Figure 5.17: Electron Beam Lithography. Procedures [10]

The electron sources vary, but the most commonly used are the thermionic source and the field electron emission source.

Lower resolution systems can use thermionic sources, which are usually formed from LaB_6 . Systems with higher resolution requirements need to use field electron emission sources, such as heated W/ZrO_2 for lower energy spread and enhanced brightness. Thermal field emission sources are preferred over cold emission sources, in spite of the former's slightly larger beam size, because they offer better stability over typical writing times of several hours.

5.3.2 Focused Ion-beam Etching

An ion beam [11] is a type of charged particle beam consisting of ions. Ion beams have many uses in electronics manufacturing and in other industries. One type of ion beam source is the duoplasmatron. Ion beams can be used for sputtering or ion beam etching and for ion beam analysis.

An ion beam can be applied for etching, feature that was explained in Section 5.2.2, it is also called sputtering, is a technique conceptually similar to sandblasting, but using individual atoms in an ion beam to ablate a target. Reactive ion etching is an important extension that uses chemical reactivity to enhance the physical sputtering effect.

In a typical use in semiconductor manufacturing, a mask is used to selectively expose a layer of photoresist on a substrate made of a semiconductor material such as a silicon dioxide or gallium arsenide wafer. The wafer is developed, and for a positive photoresist, the exposed portions are removed in a chemical process. The result is a pattern left on the surface areas of the wafer that had been masked from exposure. The wafer is then placed in a vacuum chamber, and exposed to the ion beam. The impact of the ions erodes the target, abrading away the areas not covered by the photoresist.

Focused Ion Beam (FIB) (Figure 5.18) instruments have numerous applications for characterization of thin film devices. Using a focused, high brightness ion beam in a scanned raster pattern, material is removed (sputtered) in precise rectilinear patterns revealing a two dimensional, or stratigraphic profile of a solid material. The most common application is to verify the integrity of the gate oxide layer in a Complementary Metal Oxide Semiconductor (CMOS) transistor. A single excavation site exposes a cross section for analysis using a scanning electron microscope. Dual excavations on either side of a thin bridge are utilized for preparing transmission electron microscope samples.

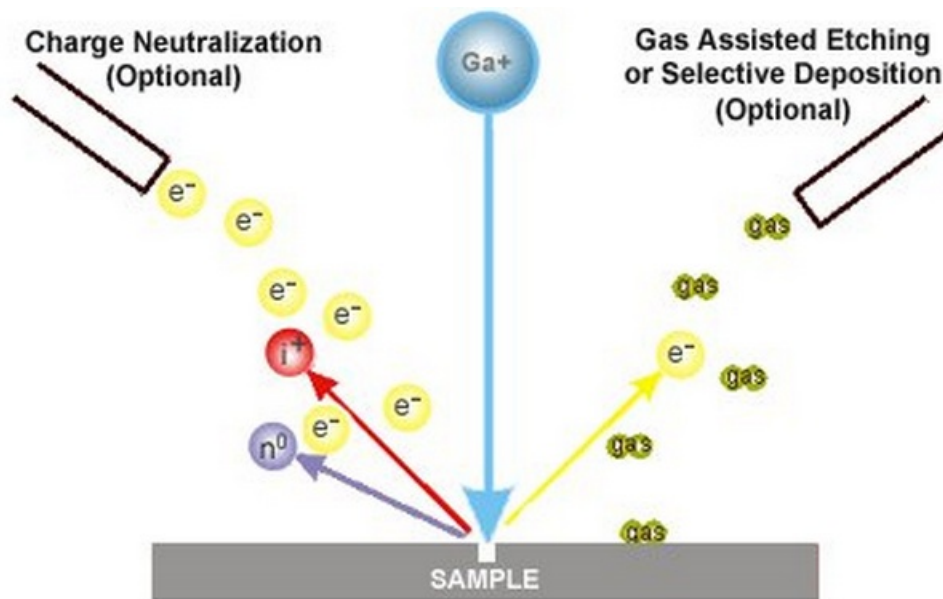


Figure 5.18: FIB Principle [11]

Another common use of FIB instruments is for design verification and failure analysis of semiconductor devices. Design verification combines selective material removal with gas assisted material deposition of conductive, dielectric, or insulating materials.

Engineering prototype devices may be modified using the ion beam in combination with gas assisted material deposition in order to rewire an integrated circuit's conductive pathways. The techniques are effectively used to verify the correlation between the Computer Aided Design (CAD) design and the actual functional prototype circuit, thereby avoiding the creation of a new mask for the purpose of testing design changes.

Sputtering is also used in materials science for extending surface analytical techniques such as secondary ion mass spectrometry or electron spectroscopy so that they can depth profile them.

Chapter 6

Design and Simulation

This Chapter explains the phases of the design and optimization of the MMI coupler, passive component that derives from the multimode waveguides, and that can be studied through the use of numerical analysis methods which are at the core of different simulation softwares, we will then determine which of the mentioned softwares is more suitable for the development of this project. Also, the description of the waveguide model already established will be given, since this will be the starting point of our design, which will be extensively detailed through its evolution. Finally, the results will be exposed and analyzed to objectively determine its advantages.

6.1 Numerical Modeling Methods

A computer simulation software [20] [21] is a computer program, that attempts to simulate an abstract model of a particular system, meaning the representation of a real-life system through the use of a high level programming language. Computer simulations has shown to offer great advantage in the field of mathematical modeling, it has been used in many systems, such as in physics, and engineering, together with other scientific areas. The system simulation can be used to explore and gain new insights into new technologies, and to estimate its performance, which can lead to the improvement, of systems, which are too complex for analytical analysis, and therefore the obtainment of any accurate solutions.

Numerical methods [22] [23] study algorithms that use numerical approximation, instead of general symbolic manipulations for the problems of mathematical analysis, as distinguished from discrete mathematics, it can be defined as a empirical method to solve a problem, meaning the approach of the unknown through the collection of data to base a theory or solution. Numerical analysis does not seek exact answers, because exact answers are often impossible to obtain in practice. Instead, much of numerical analysis is concerned with obtaining approximate solutions

while maintaining reasonable bounds on errors. Numerical analysis naturally finds applications in all fields of engineering and the physical science. Numerical stability is the most desirable property sought in numerical algorithms. Other critical characteristics of a numerical computing analysis method is the accuracy, the rate of convergence and the efficiency.

As explained previously, numerical analysis is an essential component of modern science and engineering, for the modeling of numerous technologies and theories. On the following sections we will mention a collection of software programs available for numerical simulations of waveguiding structures. The software is divided into two major categories of our concern: Mode Solvers, Wave Propagators.

6.1.1 Mode Solvers

Modes [24] are the fields that maintain the same transverse distribution and polarization at all distances along the waveguide axis. For a limited number of ideal, simple waveguide configurations, modes and corresponding propagation constants can be found analytically. However, for a majority of realistic waveguides, numerical simulations are essential. Even the waveguides with relatively simple cross-sections and refractive index profiles in reality may suffer from anisotropy, inhomogeneities due to fabrication tolerances, and material losses that affect their modal properties. Depending on the refractive index profile and other waveguide characteristics, various types of modes may be supported, including antiguiding, leaky, lossy, or radiating modes.

This section contains several mode solvers based on numerical techniques of different complexity and specialization. Some of the tools can simultaneously be used as mode solvers and as wave propagators (Transfer-Matrix Method, Eigenmode Expansion Method, Beam Propagation Method, Finite Element Method), while others are Mode Solvers only (Plane Wave Expansion Method, Multipole Method, Source Model Technique). While some of the numerical methods, such as the Finite Element Method, the Plane Wave Expansion Method, the Beam Propagation Method-based mode solvers, the Film Mode Matching Method, and the Finite-Difference Method, can be used for finding modes of the arbitrary cross-section and refractive index profile waveguides, others are more specialized. For instance, the Multipole Method (CUDOS MOF Utilities) and the Source Model Technique (developed by Professor Leviatan's group at Technion) are techniques that are particularly useful mode-finding modes of microstructured or photonic crystal fibers.

There are also methods that include several software packages that comprise more than one mode-solving technique (FIMMWAVE Mode Solvers by Photon Design, MODE Solutions by Lumerical, OlympIOs Mode Solver Modules by C2V, OptiFIBER by Optiwave). These packages include various techniques, ranging from approximate analytical methods, including the Effective Index Method and the Marcatili's Method (OlympIOs Mode Solver Modules by C2V), and the Gaussian Mode Fiber Solver (FIMMWAVE Mode Solvers), to advanced numerical methods, such as the semi-vectorial and full-vectorial Finite-Difference Method (MODE Solutions, OlympIOs Mode Solver Modules, OptiFIBER by Optiwave), the Film Mode Matching Method (FIMMWAVE Mode Solvers by Photon Design, OlympIOs Mode Solver Modules), the Transfer-Matrix Method (OptiFIBER by Optiwave), and the Finite Element Solver (FIMMWAVE Mode Solvers by Photon Design). Depending on the problem, required accuracy and simulation speed, the user can choose an appropriate method. The following methods only use a single numerical

tool, the Beam Propagation Method-based mode solvers, utilizing the iterative method and the correlation method (BeamPROP Mode Solvers by RSoft), and the Wave Matching Technique (developed by Dr. Hammer at the University of Osnabrück).

6.1.1.1 Finite Element Method (FEM)

The FEM [25] [24] methods have been implemented in both time and frequency domains, and can be used as wave propagators and mode solvers simultaneously. However, some of the FEM software packages work only as mode solvers and should not be confused with those allowing propagation effects studies (FEMSim by RSoft, FIMMWAVE-FEM-Solver by Photon Design, pdnMesh free software).

The FEM is a numerical technique for finding approximate solutions to Partial Differential Equations (PDE) and their systems as well as integral equations, that handle complex geometries, and boundaries, such as waveguides with arbitrary cross-sections, with relative ease. The field region is divided into elements of various shapes, such as triangles and rectangles, allowing the use of an irregular grid, in simple terms, it divides up a very complicated problem into small elements that can be solved in relation to each other. The solution approach is based either on eliminating the differential equation completely (steady state problems), or rendering the PDE into an equivalent ordinary differential equation, which is then solved using standard techniques, such as finite differences. In a context of optical waveguides, the FEM can be used for mode solving and propagation problems. Two approaches to solve waveguide problem include the variational method and the weighted residual (Galerkin) method. Both methods lead to the same eigenvalue equation that needs to be solved.

6.1.1.2 EigenMode Expansion Method (EMEM)

The EMEM [24] is based on a rigorous solution of Maxwell's equations, representing the electromagnetic fields everywhere in terms of a basic set of local modes. In principle an exact solution can be obtained using an infinite number of modes in our expansion. Of course, in practice the number of modes is limited and there will be numerical errors in the implementation, as in any numerical technique. To obtain higher accuracy one can simply add more modes, enabling the method to accurately compute problems that cannot be computed with other techniques. The algorithm is inherently bi-directional and utilizes the scattering matrix (S-matrix) technique to join different sections of the waveguide or to model nonuniform structures. The scattering matrix technique relates the incoming waves, i.e. the forward wave at the beginning of the section and the backward wave at the end of the section, to the outgoing waves, i.e. the backward wave at the beginning and the forward wave at the end. All reflections are taken into account in the method. If one part of a device is altered, only the S-matrix of that part needs to be re-computed. The method can simulate light propagating at any angle, even 90 degrees to the propagation axis, simply by adding more modes. Applications include diffractive elements, directional couplers, tapers, MMIs, bend modeling, periodic structures and others. Structures with a very large cross-section are less suitable for the method since computational time typically scales in a cubic fashion, with, for example, cross-section width.

CAvity Modelling FRamework (CAMFR) is a commonly used simulation program, it is a Full-vectorial Maxwell solver based on the EMEM. Although it can handle general electromagnetic problems, its main focus is on applications in the field of photonics. CAMFR treats the field as a sum of local eigenmodes in each z-invariant layer and does not rely on spatial discretisation and finite differences to solve Maxwell's equations, as other methods, like Finite-Difference Time-Domain (FDTD) 6.1.2.5, do.

Since there is no spatial discretisation, CAMFR can be orders of magnitude faster than FDTD for a large class of structures. Incorporates advanced boundary conditions, which can drastically improve simulation accuracy and speed. CAMFR is an ongoing active research project. Two-dimensional Cartesian structures and three-dimensional cylindrical symmetric structures CAMFR can be used to calculate the scattering matrix of a structure, the field inside a structure, for any given excitation, the band diagrams of an infinite periodic structure, the threshold material gain and resonance wavelength of laser modes, and the response to a current source in an arbitrary cavity.

Defining structures is quite straightforward, either layer-by-layer, or using geometric primitive shapes. There are also integrated plotting routines for rapid simulation feedback. CAMFR is conceived as a C++ framework, with all the algorithms implemented in terms of abstract waveguides and scatterers. This makes it extremely easy to extend CAMFR to new geometries. The end user does not deal with this C++ code directly, but rather through bindings to the Python scripting language. It is specialized in the simulation of wavelength-scale microstructures (like photonic crystal devices), lasers (like vertical-cavity surface-emitting lasers) and light-emitting diodes (like resonant-cavity Light Emitting Diodes (LED)s).

Another simulation tool is FIMMPROP, a highly innovative tool for simulating propagation in optical waveguides in 2D and 3D, which is fully integrated as part of FIMMWAVE, a generic full vectorial optical mode solver for waveguide structures, and relies on the rigorous EMEM.

The EMEM enables the fields to be calculated using fast semi analytical methods. The speed and nature of the calculations allow the propagation to be treated in a fully bi-directional manner, taking into account all the reflections at intermediate joints. This makes FIMMPROP capable of modeling structures which are insolvable by other methods such as BPM.

It is used in the modeling of devices with strong internal reflections, such as waveguides terminated by a tilted facet. Use of semi-analytical modal methods by FIMMPROP permits calculations to achieve high accuracy even for more complicated structures. Calculations permit both transmission and reflection coefficients of the modes at each joint to be determined for use in the fully bi-directional propagation algorithm. Structures with Z-varying cross-sections such as tapers and Y-junctions are computed using sophisticated extensions to the EMEM. The algorithm builds a scattering matrix description of the device and all its elements, which means that once the matrices are generated, the response to many different input profiles can be found without further computation, for example to obtain response to both TE and TM excitation. If the structure is altered, the routine needs only recalculate the elements that have changed.

It is used to model devices and components like the MMI couplers, Mode converters, Codirectional couplers (polarization converters), Bends, Periodic structures, Tilted joins, Facets and free space regions for waveguide-gap-fiber simulations.

6.1.1.3 Full-Vectorial Finite-Difference Method (FV-FDM)

The full-vector finite difference mode solver [24] analyses micro-structured optical fibers. It is a new mode solver, a hybrid, that is based on discretization scheme was first proposed by Yee. Yee's mesh is widely used in the FDTD analysis. In the FV-FDM the Yee's two-dimensional mesh is used for complex optical waveguides. It is widely used for the modeling of MMI couplers [26] [27]. It permits the determination of the working principle of devices and to evaluate their performance. It can be directly utilized for evaluating effective refractive index distributions, as well as eigenmode characteristics, and the calculation of effective indexes of the TE and TM polarized modes.

It shows an improvement for all of the Finite Difference mode solvers and is particularly effective when combined with Yee's mesh. This results in significantly accelerated convergence and improved numerical accuracy. The validity and effectiveness of this mode solver is the reason for its high usage in the MMI device simulation.

6.1.1.4 Full-Vector Finite-Element Beam Propagation Method (VFE-BPM)

The VFE-BPM [28], is another hybrid which uses both wave propagation analysis and mode solvers. In terms of all the components of slowly varying electric fields is described for the analysis of 3D nonlinear optical waveguides. Electric fields obtained with this approach can be directly utilized for evaluating nonlinear refractive index distributions. To eliminate nonphysical, spurious solutions, hybrid edge/nodal elements are introduced. Furthermore, to avoid spurious reflections from the computational window edges, anisotropic perfectly matched layer boundary conditions are implemented, and to reduce computational effort for the nonlinear optical waveguide analysis, an iterative algorithm is also introduced. The effectiveness of the present approach is verified by way of numerical examples: nonlinear directional couplers, spatial soliton emission phenomena, and soliton couplers.

The method is useful for analysis of a three-dimensional nonlinear optical waveguide is newly developed with all slowly varying electric field components as unknown variables. The derived electric field can be used directly for evaluation of the nonlinear refractive index, so that a nonlinear refractive index directly corresponding to the electric field can be obtained. Various methods are included for improving the performance of the VFE-BPM such as the edge/nodal hybrid elements for suppressing generation of spurious solutions, the anisotropic perfectly matched layer for prevention of spurious reflection from the analysis domain edges, and an iteration method to relax the step width in the propagation direction.

6.1.2 Wave Propagators

Wave propagation techniques [24] are essential components of waveguide and waveguide-based photonic device design. While mode-solving methods are based on the assumption that the guiding structures are uniform in the propagation direction, many practical waveguide configurations, such as tapers, bends and junctions, are actually nonuniform in the longitudinal direction. The analysis of these types of waveguides relies on propagation-type techniques. Even in

uniform structures, propagation methods provide important information about beating between modes in multimode structures, mode filtering upon propagation, and mode coupling effects. In addition, propagation methods facilitate numerical studies, accurately reflecting wave guiding properties of realistic devices and providing insight into the field's evolution inside the structure. In these simulations arbitrary launch conditions are usually allowed and the evolution of such an arbitrary beam/pulse can be monitored at any point inside the structure. This section includes both time-domain methods, including the FDTD, the Split-Step Method (SSM), and the FEM, and frequency-domain methods, such as the BPM, the Transfer Matrix Method (TMM), and the EMEM.

6.1.2.1 SSM

The SSM [24] is a Nonlinear Schrodinger Equation Solver provided by Professor Agrawal's group at the Institute of Optics , University of Rochester , OptSim by RSoft, SSPROP provided by Professor Murphy, University of Maryland , College Park. It is a simple and straightforward technique for the modeling of linear and nonlinear pulse propagation of optical fibers. It allows the inclusion of all major effects taking place in realistic optical fibers, including group-velocity dispersion and higher-order dispersion, self-phase modulation, cross-phase modulation, four-wave mixing effects, Raman amplification and many other linear and nonlinear phenomena.

6.1.2.2 TMM

The TMM [24] (GratingMOD by RSoft, OptiGrating by Optiwave, TMM Mode Solver and Anisotropic TMM Mode Solver developed by Henri Uranus at the University of Twente, WAVEGUIDE provided by the Southern Methodist University group) and the EME (CAMFR developed by Ghent University group, FIMMPROP by Photon Design, OlympIOs Bidirectional Eigenmode Propagation Module) methods are relatively simple methods that are inherently bidirectional. They are especially useful for the modeling of waveguides comprising grating structures, waveguide tapers and couplers.

6.1.2.3 BPM

The BPM [24] (implemented in BeamPROP by RSoft, OlympIOs by C2V, OptiBPM by Optiwave, OptoDesigner by PhoeniX) is a widely used and well-developed method that solves wave propagation problems for a given frequency. Several advanced techniques, including semi-vectorial and vectorial implementations, wide-angle and bi-directional BPM, have been developed, making the BPM a versatile technique for a wide class of optical waveguides with arbitrary cross-sections and complex refractive index profiles. It is the most commonly used method in MMI devices simulation, since it takes all modes into account, including radiative modes, and therefore provides the most accurate model for most optical devices.

The BPM was developed for the analysis of light propagation in slowly varying nonuniform guiding structures, such as tapers, bends, and couplers. Various implementations of BPM differ

by the method used for calculations of the derivatives with respect to the coordinates in the propagation directions and include the Finite-Difference Beam Propagation Method (FD-BPM), Fast Fourier Transformer Beam Propagation Method (FFT-BPM), and the Finite-Element Beam Propagation Method (FE-BPM). Historically, the FFT-BPM was widely used until the FD-BPM was developed. The standard form of implementation of the FD-BPM is the Crank-Nicholson method. The standard BPM solves the Helmholtz equation in paraxial approximation.

The paraxial approximation enables BPM to analyze much longer structures (on the millimeter and centimeter scale) than would be realistic for analysis with FDTD. While the BPM is usually used for the modeling of low-index-contrast waveguides, some relatively high-index contrast structures can be studied as well. Scalar, semi-vectorial, and vectorial formulations have been developed, and several advanced features have been proposed and implemented in BPM-based software packages, including wide-angle formulation and bi-directional propagation. The BPM versions that incorporate nonlinearities, anisotropy, gain and loss have been described and implemented. In addition, two BPM-based mode solvers (based on imaginary-distance BPM and correlation technique) have been proposed. Typical BPM applications include standard, specialty, and photonic crystal fibers, switches, modulators, multimode interference devices, and passive splitters.

BeamPROP [29] is the industry-leading design tool based on the BPM for the design and simulation of integrated and fiber-optic waveguide devices and circuits. The software has been commercially available since 1994, and is in use by leading researchers and development engineers in both university and industrial environments worldwide. It has been proven to be a top algorithm for fast and accurate device design. The BeamPROP also has a built-in advanced AWG utility for simplifying router and demultiplexer design. Advanced capabilities allow for the simulation of complicated devices, and it is fully integrated into the RSoft CAD Environment.

BeamPROP has applications in a wide range of integrated and fiber-optic devices including, but not limited to:

- Wavelength-Division Multiplexing (WDM) devices such as Arrayed Waveguide Grating (AWG) routers.
- Switches, like the directional coupler-based or digital-y type.
- Modulators, for example the Mach-Zehnder type
- Multimode interference devices.
- Passive $1 \times N$ or $N \times N$ splitters.
- Laser structure transverse mode analysis
- Standard and specialty fiber design.
- Gratings
- Sensor structures.

The BeamPROP delivers superior robust and efficient results based on an implicit finite-difference scheme. It has numerous features, such as:

- 2D and 3D simulation capabilities.
- Non-uniform mesh.
- Anisotropic and non-linear materials.
- Electrode and heater effects.
- Polarization effects and coupling via a full-vectorial BPM implementation.
- Wide-angle propagation can be improved by multistep Padé approximation techniques, a variable reference wave number, and conformal index mapping of bends to allow for accurate and efficient off-axis propagation.
- Bidirectional BPM formulation for considering reflection along the propagation direction to be considered.
- Two BPM-based mode-solvers for the computation of modal propagation constants and profiles for both guided and radiation modes for 2D and 3D geometries.
- Comprehensive measurement tools to compute fields, power distribution, loss, etc.

6.1.2.4 Finite-Difference Method (FDM)

The FDM [30] is a numerical methods for approximating the solutions to differential equations using finite difference equations to approximate derivatives. The FDM relies on discretizing a function on a grid. To use the finite difference method to attempt to solve (or, more generally, approximate the solution to) a problem, one must first discretize the problem's domain. This is usually done by dividing the domain into a uniform grid.

The previous explained procedure leads the finite-difference methods to produce sets of discrete numerical approximations to the derivative, which generally can create errors. This error in a method's solution is defined as the difference between its approximation and the exact analytical solution. A finite difference is a mathematical expression of the form $f(x+b) - f(x+a)$. If a finite difference is divided by $b - a$, one gets a difference quotient. The approximation of derivatives by finite differences plays a central role in finite difference methods for the numerical solution of differential equations, especially boundary value problems. The two sources of error in finite difference methods are round-off error, the loss of precision due to computer rounding of decimal quantities, and truncation error or discretization error, the difference between the exact solution of the finite difference equation and the exact quantity assuming perfect arithmetic.

6.1.2.5 FDTD Method

The FDTD [31] [24] is a numerical analysis technique used for modeling computational electrodynamics, it is implemented in a large number of free and commercially available packages, it is the most general and rigorous time-domain method, and because of this characteristic this system is fit to cover a wide range of frequencies with just one simulation run. It is particularly useful for a large number of optical guiding structures, such as photonic crystal waveguides, surface plasmon waveguides, high-index contrast waveguide devices, ring and disk resonators, negative index material structures, dispersive, and nonlinear materials since it tries its properties in a natural way. Modal properties can be extracted after a single propagation. However, the FDTD may become prohibitively computationally intensive and therefore impractical for long waveguides and devices.

The FDTD technique is based on a discrete representation of time-dependent Maxwell's equations on a grid that is exact in the limit that the grid spacing goes to zero. Various optical materials can be used with FDTD algorithms. Current FDTD modeling applications range from near-DC (ultralow-frequency geophysics involving the entire Earth-ionosphere waveguide) through microwaves (radar signature technology, antennas, wireless communications devices, digital interconnects, biomedical imaging/treatment) to visible light (photonic crystals, nanoplasmonics, solitons, and biophotonics).

Modal properties can be extracted in a single simulation by Fourier-transformation of the time-varying response of the system to some input. Then, the peaks in the resulting spectrum correspond to the eigenfrequencies. The primary limitation of FDTD is that it is computationally intensive, especially in the three-dimensional case, but the method is readily applicable to the two-dimensional structures. In practice, FDTD needs to have at least 10 to 20 cells per minimum wavelength in every direction, depending on the application and accuracy required. Also, even higher sampling rates may be required to reduce the cumulative numerical dispersion error, which is proportional to the length of wave propagation. This renders FDTD impractical for large devices.

Various techniques have been proposed to make the FDTD method more efficient, including pseudospectral time-domain techniques, nonuniform and adaptive mesh refinement, hybrid FDTD-Finite Element techniques, and hardware acceleration concepts. Different software providers take various approaches to optimize the performance of their software in solving problems that require clustering. The approaches include parallel processing features in the software and, most recently, dedicated hardware to increase single system processing speeds. Typical applications for optical software based on an FDTD algorithm include the modeling of high-index-contrast waveguides, photonic-bandgap structures for photonic crystals, ring resonators, nano-plasmonic devices, gratings and other diffractive structures, dispersive, nonlinear and gain materials, and biophotonics.

6.2 Waveguide Model

Fraunhofer IZM with previous work has already established a general structure for the components in its optical circuits, and therefore the passive component that will be developed in this

project shall follow this basic specifications. The description given will detail both for Single mode and for Multi mode waveguides the layer materials and primary dimensions. This will allow its future incorporation into more complex circuits with as much simplicity as possible. For the same reasons all designs where made in 3D, despite the fact that these devices have the same behavior through the whole transversal direction, permitting the obtainment of similar results with a 2D design.

6.2.1 Single Mode Waveguide

In Chapter 2 the waveguide characteristics where exposed and detailed since the design is based on the SOI slab waveguide with a ridge etched into the upper substrate,changing the effective indices of refraction on either side, creating conditions for Total internal reflection (TIR) to occur within the ridge area allowing for guidance of light. The materials to be used are shown in Table 6.1 together with their respective index, and the dimensions and arrangement of each layer in Figure 6.1 .

Material	Refractive Index
Air	1
Silicon	3.48
Silicon Dioxide	1.46

Table 6.1: Materials and Refractive index of the Optical Waveguide

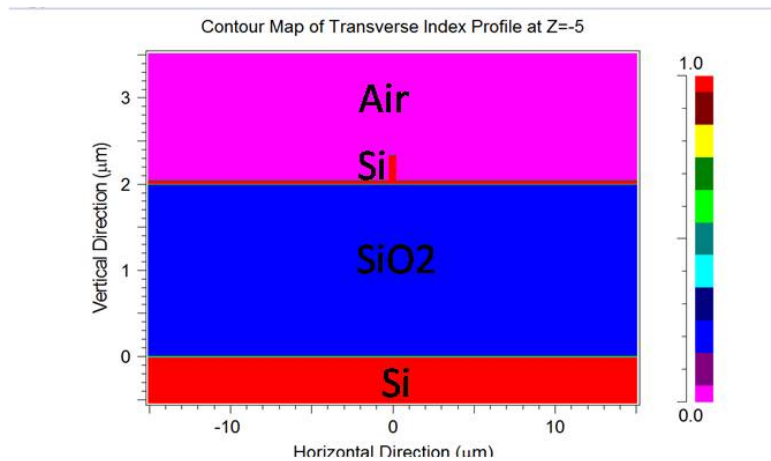


Figure 6.1: Single Mode Cross Section of the Waveguide

Where through a vetical cut we will have by order the Cover of the wave guide is the *Air* theoretically infinite, followed by the Ridge and Slab composed by Si (Silicon) that measure $290nm$ and $50nm$ respectively. Afterwards we encounter the Insulator formed by SiO_2 (Silicon Dioxide) which, as its name dictates, to isolate the Slab it must have a large vertical dimension therefore measuring $2\mu m$. Finally the waveguide ends with the Silicon substrate layer which is also theoretically infinite.

Horizontally the only limited layer is the Ridge which will be of $400nm$ and will be confined laterally and in the upper surface by the Air as shown in the mentioned figure.

The Single mode waveguides represent the inputs and the outputs (Figure 6.2) of the MMI device, and initially it will be used uniquely for this purpose.

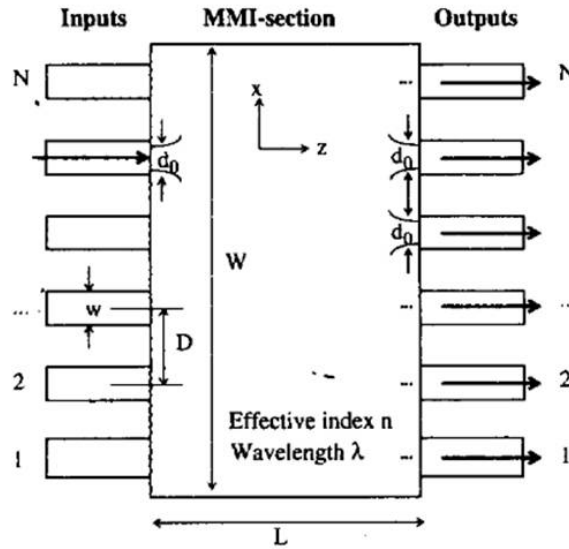


Figure 6.2: Design Layout. Input Single mode Waveguides, MMI Section, Output Single mode Waveguides [4]

6.2.2 Multi Mode Waveguide

The Multi Mode Waveguide represents the MMI section of the device shown in Figure 6.2 . Its difference with the Single Mode Waveguide relies only in the width of the Ridge since this characteristic will determine its Multi Modal properties. Its cross section is shown in Figure 6.3.

In respect to the materials used as seen in the previously mentioned figure it remains equal to the ones used in the Single Mode Waveguide, as the vertical dimensions are maintained the same. The width of the Ridge is arbitrary but to obtain a optimized result simulations must be performed keeping in mind that for the Multi Mode characteristics to prevail the width of this layer must not decrease from a certain point.

6.3 Design and Simulation

The MMI Coupler to be design will only cover, from the basic cases of self-imaging which we can observe in Figure 6.4, the Paired and Symmetric interference mechanism, leaving out the General interference case.

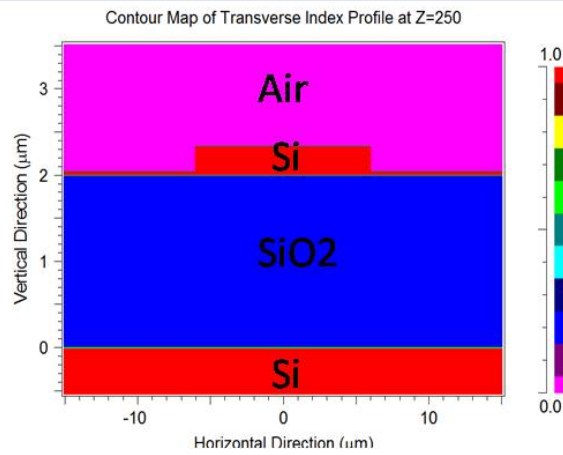


Figure 6.3: Multi Mode Cross Section of the Waveguide

This cases follow the formulas encountered in the figure and in Chapter 4. We can see the summary of analytical characteristics in Table 6.2 . From the mentioned cases we will design and simulate the 1×2 , the 1×3 , and the 2×2 coupler (Figures 6.5, 6.6 and 6.7).

Interference Mechanism	General	Paired	Symmetric
Inputs x Outputs	$N \times N$	$2 \times N$	$1 \times N$
First single image distance	$3L_\pi$	L_π	$(3L_\pi)/4$
First N fold image distance	$(3L_\pi)/N$	$(L_\pi)/N$	$(3L_\pi)/4N$
Excitation requirements	none	$c_v = 0$ for $v = 2, 5, 8, ..$	$c_v = 0$ for $v = 1, 3, 5, ..$
Input(s) location(s)	any	$y = \pm W_e/6$	$y = 0$

Table 6.2: Summary of Characteristics of the General, Paired and Symmetric Interference Mechanisms

6.3.1 Initial Design

The initial design of the MMI Coupler shall follow the standards detailed previously, but the definition of the cross section for practical reasons will be modified as shown in Figure 6.8, eliminating the last silicon layer, which will not change the result, since the Silicon Dioxide substrate layer has the task of avoiding any influence of this mentioned layer with the function of the MMI device.

To do such a modification the global description of the waveguides in the BeamPROP software simulator must be performed as shown in Table 6.3.

Our waveguide is designed in 3D therefore we must define a structure type in the description as well, filling the Layer Table 6.4 as shown.

Afterwards, the MMI sections width (W_{mmi}) is arbitrarily determined, and since the width and the length (L_{mmi}) of this segment are related through the beat length (L_π) as noted in Equation 4.5, Figure 6.4, and the next Equations 6.1 and 6.2 .

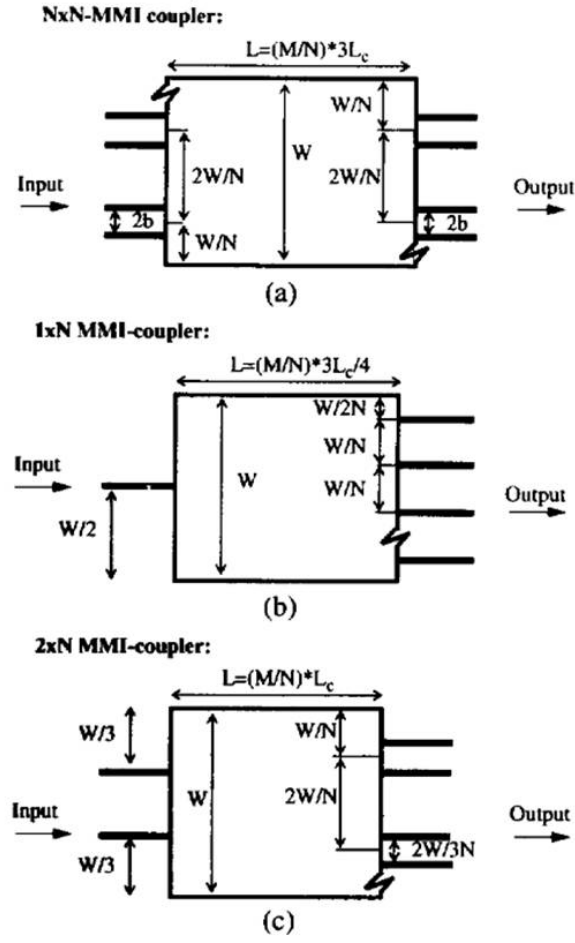


Figure 6.4: Interference Mechanisms of the MMI Couplers

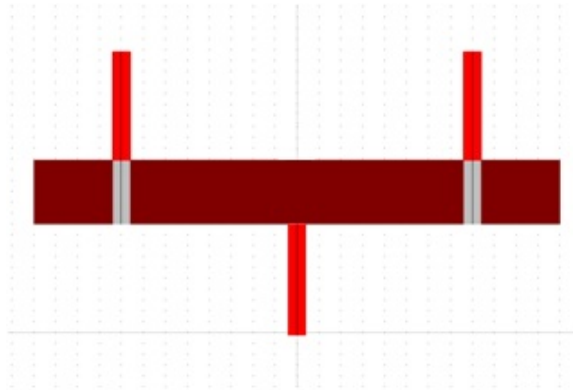


Figure 6.5: 1x2 MMI Couplers

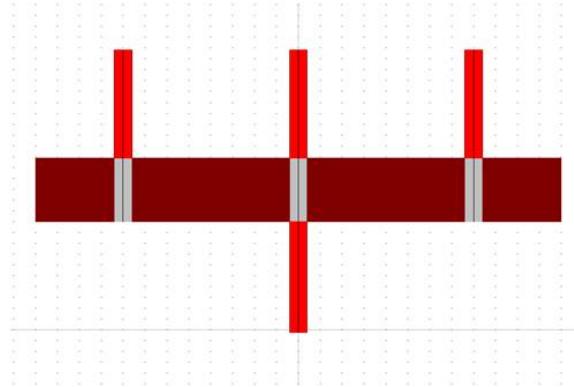


Figure 6.6: 1x3 MMI Couplers

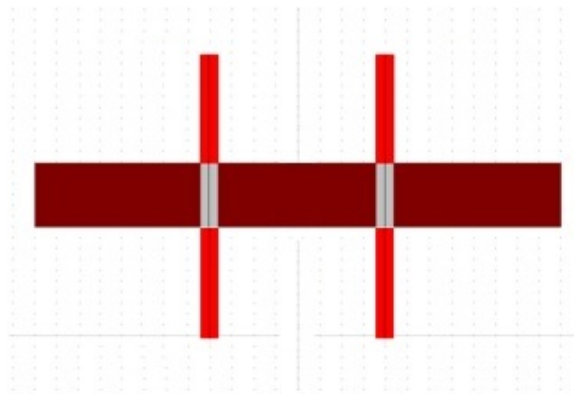


Figure 6.7: 2x2 MMI Couplers

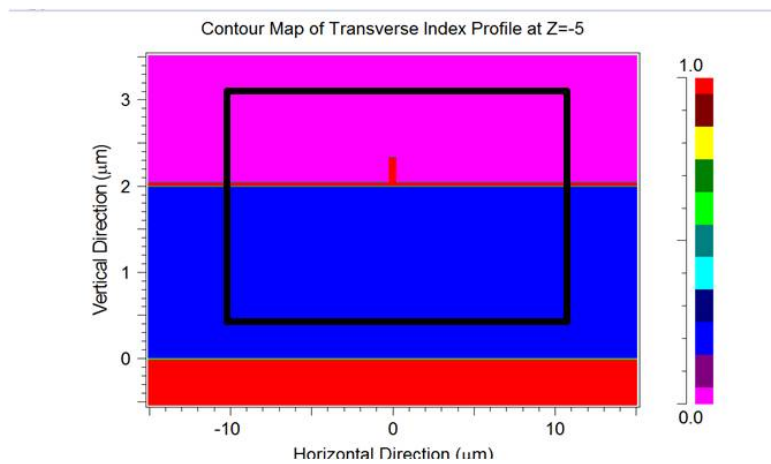


Figure 6.8: Modified Layer Structure of the Waveguide

Parameter	Value
Index Difference	2.02
Background Index	1.46
Free Space Wavelength (μm)	1.55
Waveguide Width (μm)	0.4
Waveguide Height (μm)	0.34
Cover Index	1
Slab Height (μm)	0.05

Table 6.3: Parameters of the Global Settings of the Waveguides in the BeamPROP Software Simulator

Parameter	Height (μm)	Index
Cover	∞	1
Layer 2	0.29	3.48
Layer 1	0.05	3.48
Substrate	∞	1.46

Table 6.4: Parameters for the Layer Table definition

$$L = \left(\frac{M}{N}\right) 3 \frac{L\pi}{a} \quad (6.1)$$

Where

$$a = \begin{cases} 1 & \text{for } NxN \\ 4 & \text{for } 1xN \\ 3 & \text{for } 2xN \end{cases} \quad (6.2)$$

Beginning with the verification of the excitation of the wanted modes we simulated the 1x3 MMI Coupler, obtaining the expected results of a Symmetric Interference, the excitation of modes 1, 3, 5,... (Figures 6.9, 6.10 and 6.11 respectively). This was performed by the use of the Compute Mode feature in BeamPROP. Then to run the simulation certain specifications had to be specified, such as:

Launch Field Field where as its name states, determines the point of the structure where the input field should be launched, being this the Single Mode Input Waveguide. And from all the possible fields we use a Gaussian Beam.

Monitor This will measure the desired parameter, in our case the amplitude of the field, through a defined Path.

Path To determine this, we select the Input and Output, and a inactive segment with the output that travels through the theoretical position in with the mode will be excited as shown in Figures 6.6 and 6.12.

Polarization Which would be TE or TM. Both must be simulated.

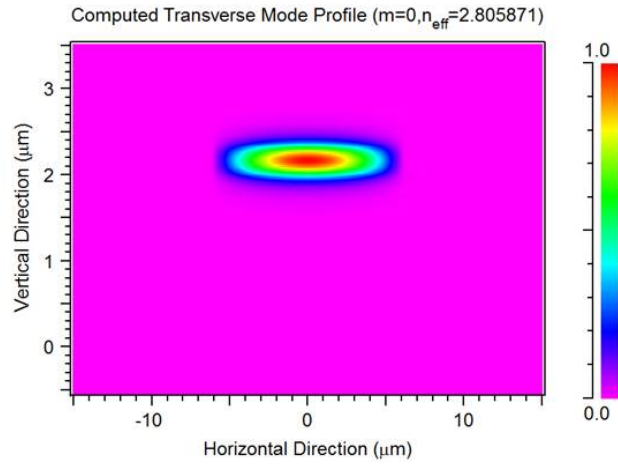


Figure 6.9: mode 1 of the Waveguide

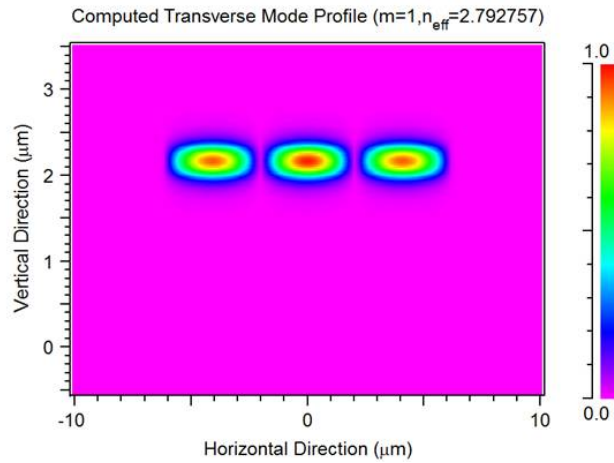


Figure 6.10: mode 3 of the Waveguide

By simulating the device with a arbitrary width (W_{mmi}) of $12\mu m$ we obtain that the length of the MMI section is in both cases very similar to the calculated by analytical methods, being the TM case practically equal. But the difference between the length of the MMI section for the TE (Figure 6.13) polarized and the TM (Figure 6.14) polarized can not be ignored, this difference is a demonstration of the polarization dependency of the structure, characteristic that requires to be modified and eliminated.

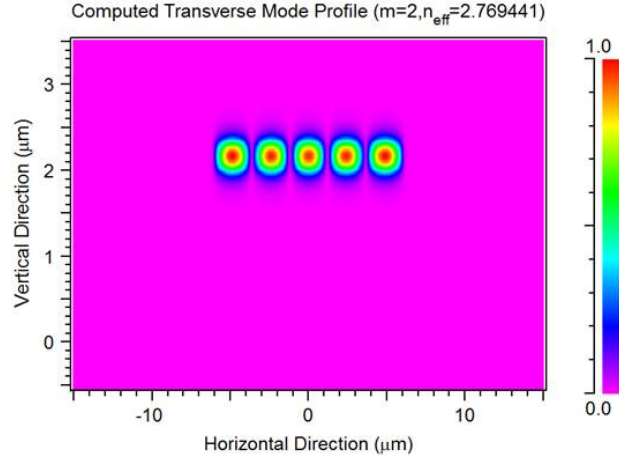


Figure 6.11: mode 5 of the Waveguide

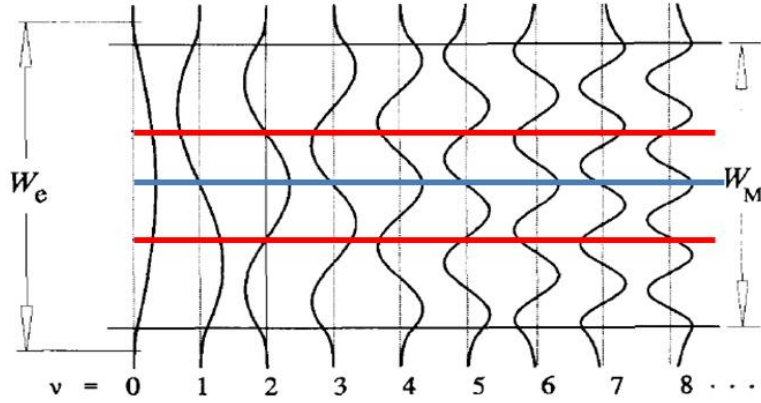


Figure 6.12: mode excitation and its amplitude lateral field profiles [4]

6.3.2 Polarization Independent Design

The encounter with a polarization dependent structure leads to take a careful look into the dimensions of the mentioned waveguide, meaning the ridge's height (h) and width (W_{mmi}), so that then the length of the MMI will be modified as well. The main objective is to obtain equal length beats, calculated with the two lowest order modes, for both polarizations (TE and TM), this is a result of different propagation constant spacing of the fundamental and the first order mode of the Multimode section.

$$\Delta L_{\pi} = L_{\pi}^{\text{TE}} - L_{\pi}^{\text{TM}} \quad (6.3)$$

One proposed solution [26] is based on the fact that ΔL_{π} is a function of the core's, meaning the ridge's, dimensions. It is proposed to modify the height of the ridge (h) to make $\Delta L_{\pi} = 0$,

Cases	L_{mmi} (μm)
Analytical method	110.864
TE Polarization	107.9
TM Polarization	110.88

Table 6.5: Lengths of a Polarization Dependent MMI Coupler

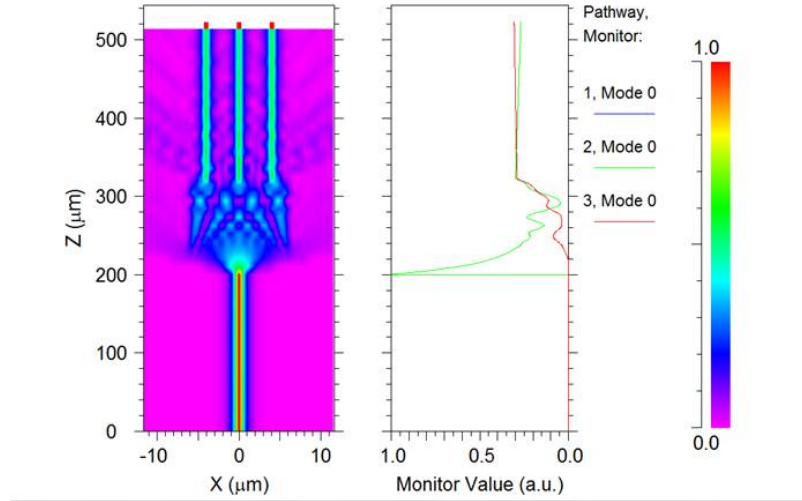


Figure 6.13: Result of the TE 1x3 MMI Coupler simulation

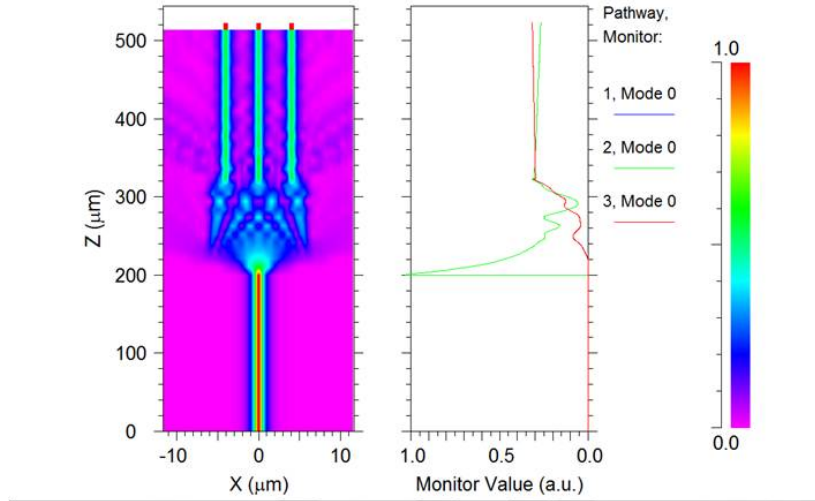


Figure 6.14: Result of the TM 1x3 MMI Coupler simulation

this is based on the characteristics of the TM modes that are sensitive to the waveguide's height due to the strong confinement to the major component of the electric field in the vertical direction

(y -direction) while the characteristics of the TE modes are not so sensitive to the height, since the major component of the electric field is in the lateral direction (x -direction). But as we recall in Table 6.6 the polarization that requires modification to achieve independence is the TE polarization, and therefore instead of varying the height of the ridge to with the TM is sensitive and the TE polarization is not, we will modify the width of the ridge, where the TE polarization is sensitive due to the strong confinement of its major electric field component until we obtain a $\Delta L_\pi = 0$.

This solution has also been taken into account in another paper [12] , with successful results (Figure 6.15) where they state that by choosing a suitable core width the birefringence in the slab waveguide will be balanced, and hence the polarization insensitive MMI coupler is achieved.

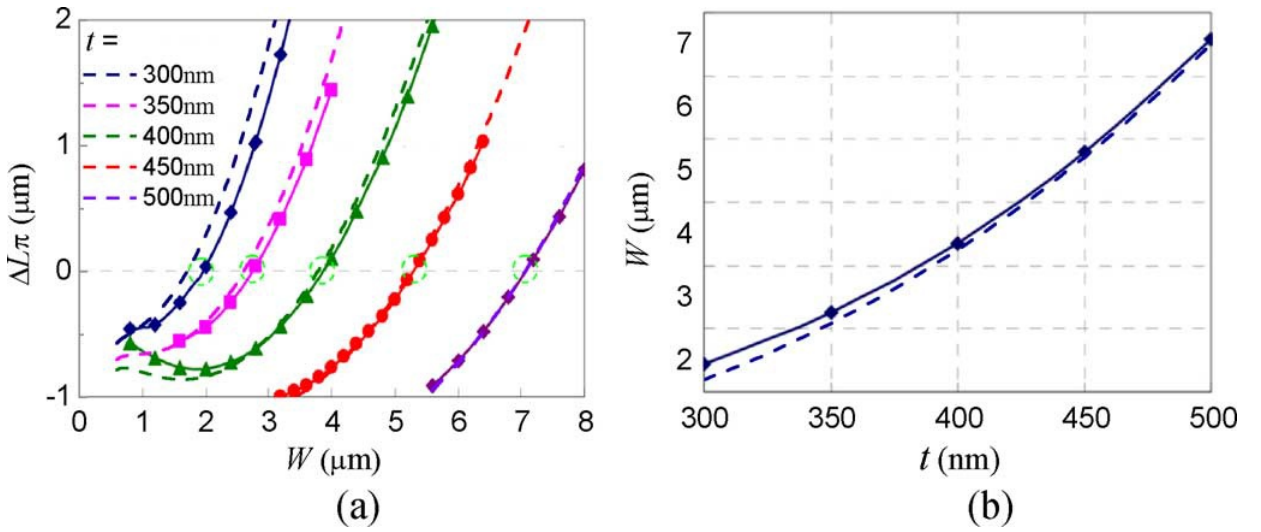


Figure 6.15: (a) Variation of beat length difference vs. core width. (b) Core width required to obtain $\Delta L_\pi = 0$ vs. height (t) [12]

When simulated we obtain a polarization independent MMI coupler, the successful result was confirmed after the collection of data from many simulations where we varied the width of the MMI section and measured the corresponding length. The results of each simulation run are shown in Table 6.6, and in Figure 6.16 permitting a easier comparison with the results in Figure 6.15 (a) from the paper that proposed this solution.

As we can see the best result is $2.5 \mu\text{m}$ but after a more fine simulation the optimum width was of $2.55 \mu\text{m}$. Also, we can appreciate in the table that the result from the simulations differ from slightly from the theoretical prediction, this is due to the approximation of the self-imaging principle explained in Chapter 4 . This width results are applicable to all the MMI Couplers to be design, since it depends on the dimensions of the ridge, not on the number of inputs and outputs.

6.3.3 Tapered Design

Though the problem with the polarization dependence was solved, another issue that requires attention is still present, and it is at this point where a general view of all 3 designs must end,

W_{mmi} (μm)	L_{mmi}^{TE} (μm)	L_{mmi}^{TM} (μm)	Theoretical L_{mmi} (μm)	ΔL_{π} (μm)
2	3	2.79	3.079	-0.21
2.5	4.298	4.3	4.811	0.002
3	6.06	6.18	6.929	0.12
4	30.58	30.94	12.318	0.36
6	42.9	44.6	27.716	1.7
10	61.96	67.6	76.989	5.64

Table 6.6: Lengths of a Polarization Dependent MMI Coupler

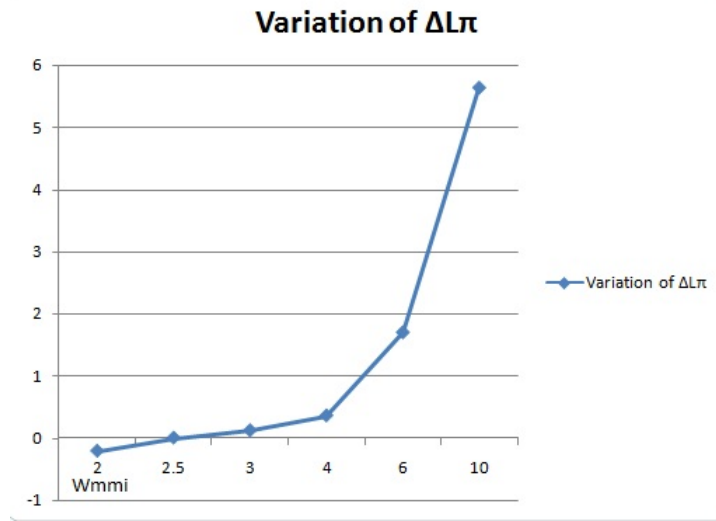


Figure 6.16: Variation of beat length difference vs. core width

since it is only in the 1x3 MMI Coupler design where we encounter this discrepancy, the presence of Cross-talk (Figure 6.17 and 6.18) between the different output SingleModed waveguides, consequence of the small size of the device's width, that creates a closeness that produces the leakage of power from one segment to another, mainly from the central output waveguide to the lateral ones. In the 1x2 MMI this does not occur, and in the 2x2 MMI it can be solved in the splitting application without the use of tapered widths, but for the switching application, tapering must be used as well, since by the use of an parabolically or exponentially tapered section we may increase the distance between the outputs and thus eliminate the coupling (cross-talk) between them.

An exponentially tapered width, as the one applied to the 1x3 MMI Coupler, shown in Figure 6.19 , and a parabolical one (Figure 6.20) used on the 2x2 switch MMI Coupler, are useful not only for the mentioned reason, but also it permits the compactness of the device through the shortage of the MMI sections length. In addition tapering the width increases the light intensity near the dielectric corners, with abrupt discontinuities, becomes smaller, and consequently the reflection is reduced. By all these means the increasing of the tapered width can improve the quality of self-imaging, and thus reduce the Excess Loss (EL).

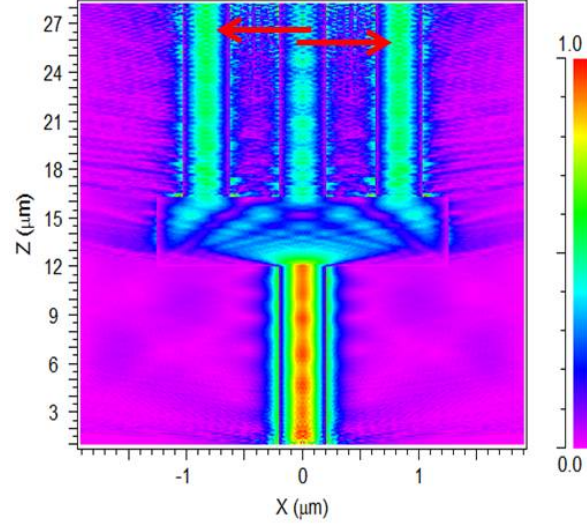


Figure 6.17: Cross-talk of the MMI Coupler's outputs

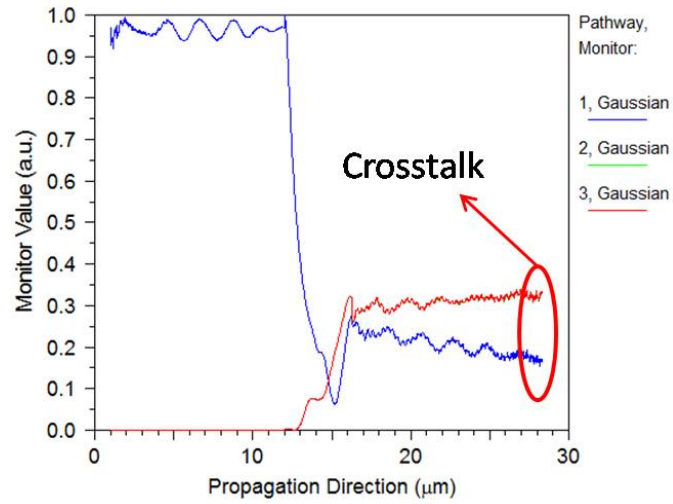


Figure 6.18: Cross-talk of the MMI Coupler's outputs. Amplitude profiles

When simulated the results obtained improved from the cross-talk perspective (Figure 6.21), but the existence of high loss and unbalance between the outputs was still a present problem, which was shared with the 1x2 MMI coupler and the 2x2 MMI coupler applied for splitting power.

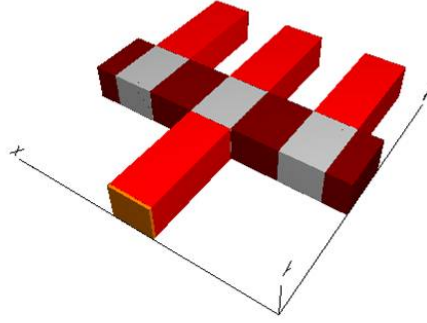


Figure 6.19: Exponentially Tapered 1x3 MMI Coupler

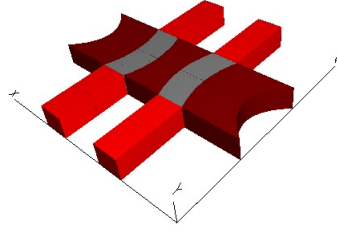


Figure 6.20: Parabolically Tapered 2x2 MMI Coupler

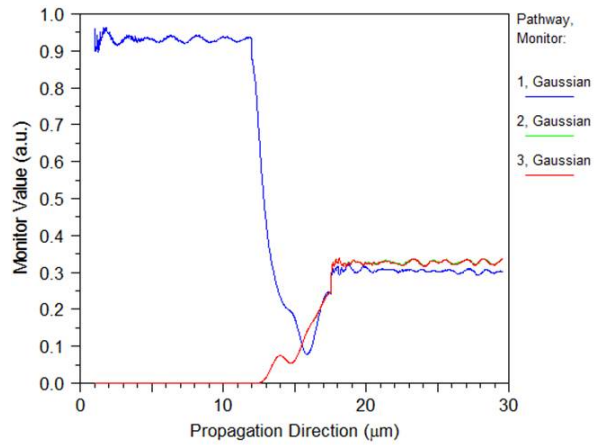


Figure 6.21: Exponentially Tapered 1x3 MMI Coupler. Imbalanced Amplitude profile

6.3.4 Low-loss Balanced Design

The use of tapered widths does not bring all these solutions without a collateral modification, we can notice that as the MMI section is widened towards the outputs, the output image is also increased in width, and for this reason singularities occur and hence discontinuities (Figure 6.22)

creating loss in the output.

The imbalance [32] [33] and excess loss in MMI couplers is primarily determined by the way the input power is partitioned into the MMI section modes. Because of this and the facts explained in the previous Section 6.3.3 we introduce linear width tapers between the MMI section and the input-output waveguides (Figure 6.23), called access waveguide, with a taper length long enough to be adiabatic, meaning that it provides an acceptable transition, this improves the image quality further making the imaged field match well with the fundamental mode of the output waveguides.

The partitioning previously mentioned is in turn determined by the normalized access waveguide width and position. There exists a number of optimum access waveguide widths that give a minimum imbalance. In particular, paired interference couplers are found to have better imbalance performance.

The imbalance between 2 outputs is determined by Equation 6.4, and the EL by Equation 6.6 .

$$imbalance = 10 \log_{10} \left(\frac{P_1}{P_2} \right) \quad (6.4)$$

$$EL = 10 \log_{10} \left(\frac{P_1 + P_2 + P_3 + \dots}{1} \right) \quad (6.5)$$

It is recommended [34] , so that the results obtained have a level of requirement, to consider only acceptable the $EL < 0.5$ dB, and in respect to the imbalance, which is certain cases has a predominant importance, for example in the MZI, to be < 0.2 dB. The results obtained without the use of the linear taper access waveguide are the following (Table 6.7 for the TE polarization and Table 6.8 for the TM polarization).

Input Output	P_1	P_2	P_3	Imbalance TE (dB)	EL_{TE} (dB)
1x3	0.241	0.282	0.241	0.682	1.169
1x2	0.413	0.413	-	0	0.83
2x2 _{switch}	0.723	0.005	-	-	1.37
2x2 _{split}	0.404	0.375	-	0.323	1.084

Table 6.7: Imbalance and EL of a width tapered MMI Coupler. TE polarization

Input Output	P_1	P_2	P_3	Imbalance TM (dB)	EL_{TM} (dB)
1x3	0.309	0.313	0.309	0.055	0.31
1x2	0.479	0.479	-	0	0.186
2x2 _{switch}	0.889	0.006	-	-	0.481
2x2 _{split}	0.491	0.491	-	0	0.078

Table 6.8: Imbalance and EL of a width tapered MMI Coupler. TM polarization

We can see that all the TE polarization results have a very elevated EL, over the level of standards, when on the TM polarization the results are very satisfactory, an so we take the solution previously proposed of the implementation a access waveguided.

As this problem was present in all the designs, the 1x3 MMI coupler, the 1x2 MMI coupler, and the 2x2 MMI coupler applied for splitting power and for switching, the linear tapered access waveguide solution was applied to all of them.

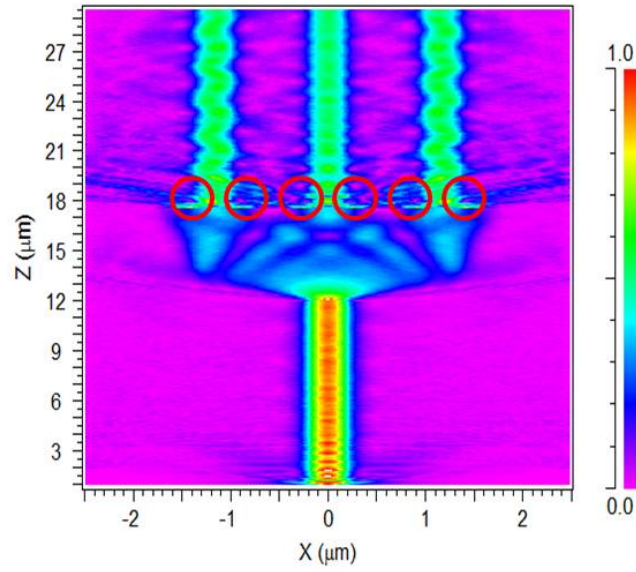


Figure 6.22: 1x3 MMI Coupler. Discontinuities

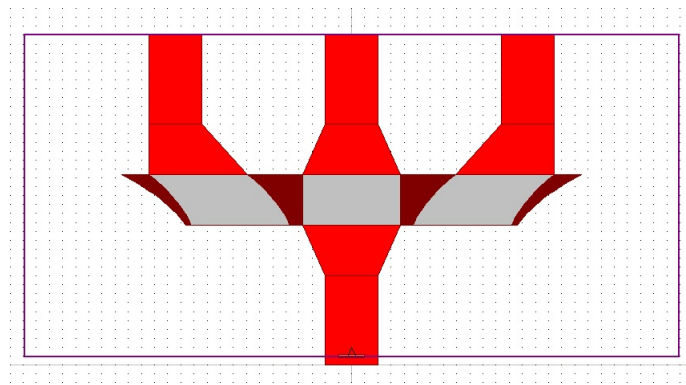


Figure 6.23: 1x3 MMI Coupler with linear tapered access waveguides

When simulated the result is satisfactory as shown in the following section, together with the ones from the rest of the designs.

6.4 Final Results

Besides all the important parameters to be defined mentioned in the previous section, we have one that is of extreme value to the MMI Coupler, since its one of its advantages against other couplers that could perform the same tasks, this parameter called the Fabrication Tolerance (FT)), determines the marge of error that the device, as its name states, tolerates for each dimension, in this case the width and the length of the MMI section. In addition the Polarization-Dependent Loss (PDL) parameter (Equation 6.6) is also calculated for utility reasons in certain applications and its the difference between the EL obtained in each polarization.

$$PDL = |EL_{TM} - EL_{TE}| \quad (6.6)$$

The following figures show the designs of all the MMI Couplers (Figure 6.24).

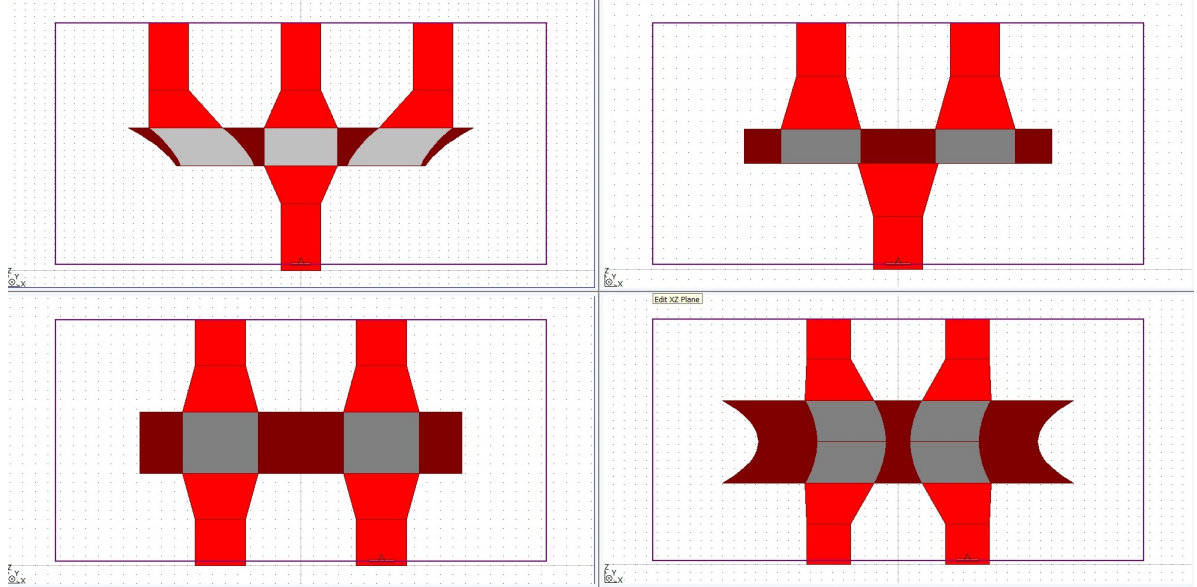


Figure 6.24: MMI Coupler Designs. 1x3(up-left), 1x2(up-right), 2x2_{split} (down-left), and 2x2_{switch} (down-right)

We can observe in the figure that both width tapered couplers, 1x3 exponentially, and 2x2_{switch} parabolically, require that the linear tapered access waveguides to have a similar angle to the tapering to avoid the presence of more discontinuities. In the 2x2_{split} coupler design the position of the inputs and outputs were modified, by separating them we avoided the cross-talk between the outputs and we obtained the images at the output in the same position as the inputs without major disruptions.

Also we observe a difference in many parameters on the design, some obtained by result of the simulations optimization (for example L_{mmi} and W_{mmi}) , and others like the tapered width of the MMI section (W_{mmi1}), and the width (W_{tap}) and length L_{tap} of the linear tapered access

waveguide were determined by running simulations in with they where varied until the most optimum result was obtained, these are:

$$1x3 \left\{ \begin{array}{l} L_{mmi} = 5.66\mu m \\ W_{mmi} = 2.53\mu m \\ W_{mmi1} = 3.5\mu m \\ L_{tap} = L_{mmi} \\ W_{tap} = 0.75\mu m \end{array} \right. \quad (6.7)$$

$$1x2 \left\{ \begin{array}{l} L_{mmi} = 6.48\mu m \\ W_{mmi} = 2.5\mu m \\ L_{tap} = 10\mu m \\ W_{tap} = 0.65\mu m \end{array} \right. \quad (6.8)$$

$$2x2_{switch} \left\{ \begin{array}{l} L_{mmi} = 20.705\mu m \\ W_{mmi} = 2.55\mu m \\ W_{mmi1} = 3.2\mu m \\ L_{tap} = 10.345\mu m \\ W_{tap} = 0.63\mu m \end{array} \right. \quad (6.9)$$

$$2x2_{split} \left\{ \begin{array}{l} L_{mmi} = 26.603\mu m \\ W_{mmi} = 2.55\mu m \\ L_{tap} = 10\mu m \\ W_{tap} = 0.6\mu m \end{array} \right. \quad (6.10)$$

The use of tapered widths waveguides produces changes in the dimensions of the device, for example the length of the MMI section, as we can observe in the previously delivered results.

The final result Figures 6.25, 6.26, 6.28 and 6.27 are shown ahead, and result tables are shown next (Table 6.9 and 6.10).

Input Output	P_1	P_2	P_3	Imbalance TE (dB)	EL _{TE} (dB)
1x3	0.309	0.313	0.309	0.056	0.310
1x2	0.465	0.465	-	0	0.315
2x2 _{switch}	0.874	0.002	-	-	0.584
2x2 _{split}	0.461	0.446	-	0.143	0.423

Table 6.9: Imbalance and EL of a width tapered MMI Coupler. TE polarization. Final Result

The PDL is determined after applying the formula exposed previously and shown in Table 6.11, together with the FT (Table 6.12) of each dimension of the MMI section, with is obtained after the parametrical simulation of the MMI coupler, being the parameter to be varied the

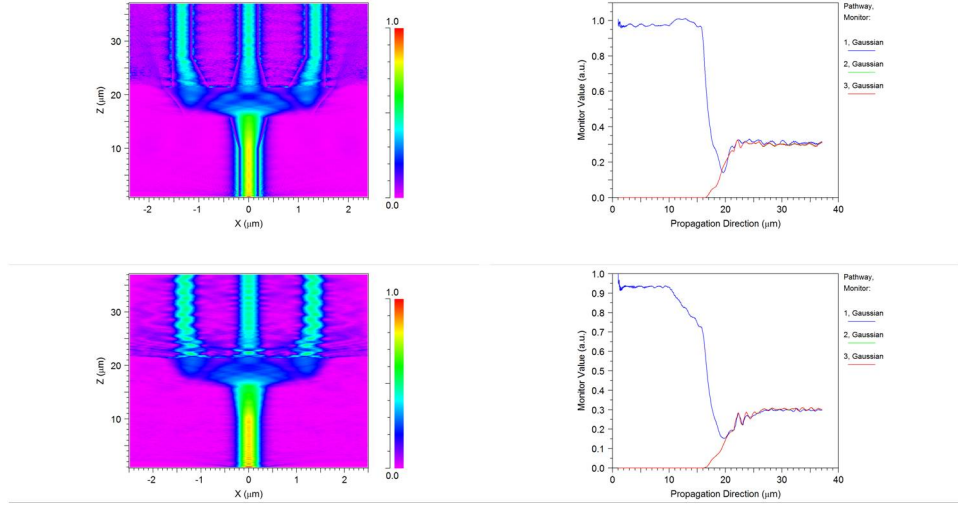


Figure 6.25: 1x3 MMI Coupler Results. TE x-z plane(up-left), TE Amplitude profile(up-right), TM x-z plane (down-left), and Amplitude profile (down-right)

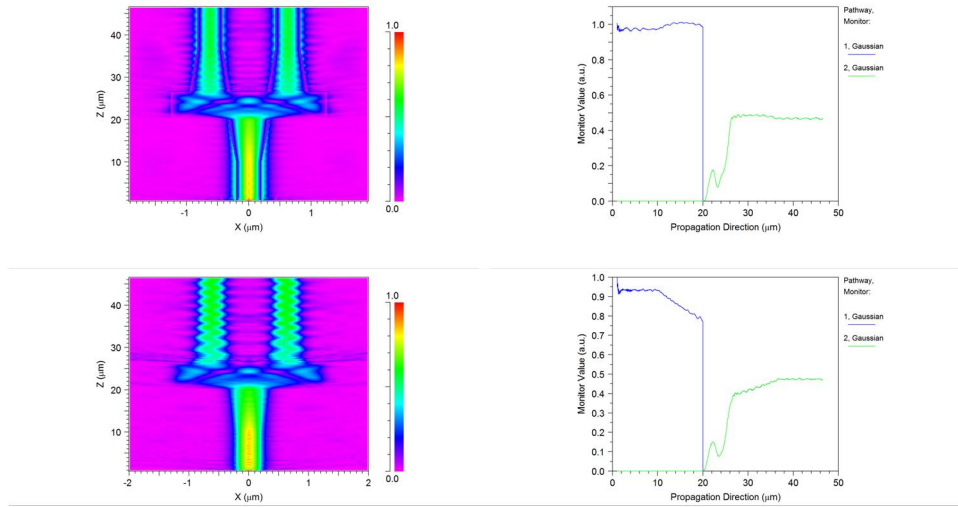


Figure 6.26: 1x2 MMI Coupler Results. TE x-z plane(up-left), TE Amplitude profile(up-right), TM x-z plane (down-left), and TM Amplitude profile (down-right)

width and the length, respectively for each case, and taking into account that the condition of minimum requirements ($EL < 1$ dB) is fulfilled.

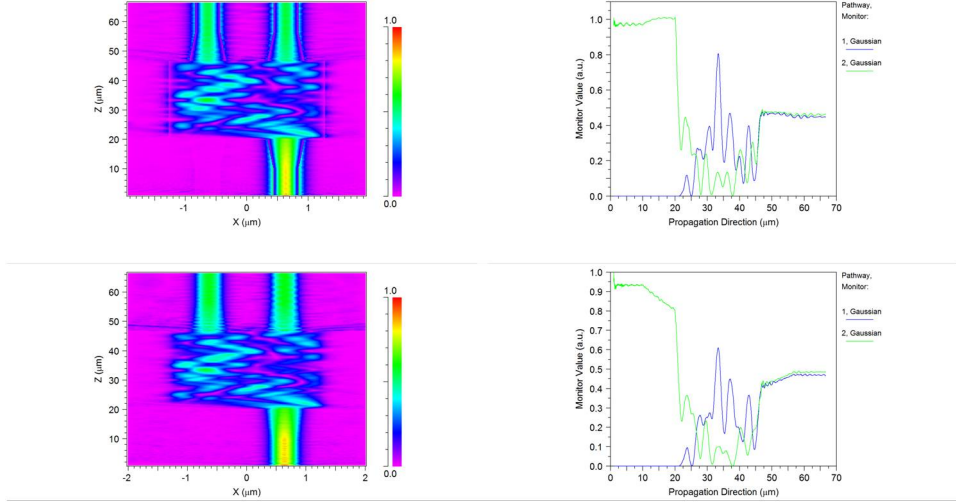


Figure 6.27: 2x2 Split MMI Coupler Results. TE x-z plane(up-left), TE Amplitude profile(up-right), TM x-z plane (down-left), and TM Amplitude profile (down-right)

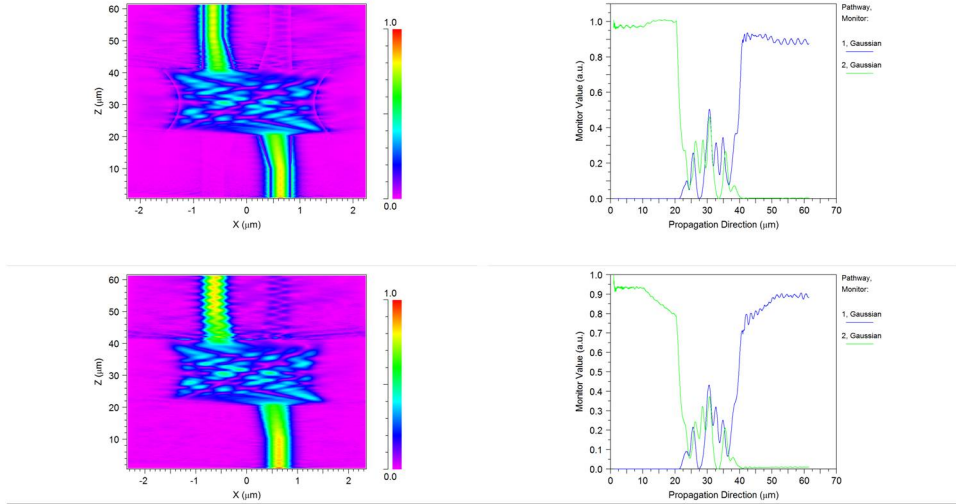


Figure 6.28: 2x2 Switch MMI Coupler Results. TE x-z plane(up-left), TE Amplitude profile(up-right), TM x-z plane (down-left), and TM Amplitude profile (down-right)

Input Output	P_1	P_2	P_3	Imbalance TM (dB)	EL_{TM} (dB)
1x3	0.303	0.298	0.302	0.072	0.438
1x2	0.469	0.469	-	0	0.278
2x2 _{switch}	0.884	0.009	-	-	0.491
2x2 _{split}	0.485	0.47	-	0.136	0.199

Table 6.10: Imbalance and EL of a width tapered MMI Coupler. TM polarization. Final Result

Input Output	PDL (dB)
1x3	0.161
1x2	0.037
2x2 _{switch}	0.093
2x2 _{split}	0.223

Table 6.11: PDL of the MMI Couplers. Final Result

Input Output	FT_{width}^{TE} (nm)	FT_{length}^{TE} (nm)	FT_{width}^{TM} (nm)	FT_{length}^{TM} (nm)
1x3	± 46	± 110	± 17	± 95
1x2	± 100	± 620	± 150	± 720
2x2 _{switch}	± 33	± 374	± 25	± 300
2x2 _{split}	± 18	± 331	± 39	± 193

Table 6.12: Fabrication Tolerance of the MMI Couplers. Final Result

Chapter 7

Applications of the MMI Coupler

The MMI Coupler has many applications, it is one of the most demanded components in the photonic industry, further applications concentrated more on making use of MMI couplers as constituents of larger passive and/or active structures. The following examples illustrate their compatibility to many types of materials and technologies, and highlight how their performances are used to an advantage.

Some of the devices in which they are used have already been mentioned in the development of this project, but in this Chapter we will explain more thoroughly the role that the MMI device plays in the most important of applications in which they are used.

7.1 Coherent Receiver Front-End

The earliest reported application using an MMI device in a complex photonic integrated circuit is the coherent receiver front-end [4] (Figure 7.1). This chip contains an MMI 3-dB coupler to combine the optical powers from the photo-signal and the local oscillator laser. In addition, two pairs of polarization-sensitive photo-detectors generate a polarization-insensitive Intermediate Frequency (IF) signal using diversity architectures. Several key features of MMI devices enhanced the overall performance of the chip.

The ultra-compact size of the MMI 3-dB coupler and its compatibility with deeply etched waveguides allowing for Single-mode ultra-compact bends. Balanced operation is important to fully use the available optical powers of the photo-signal and the Local Oscillator (LO) laser and to minimize common mode noise due to LO intensity fluctuations.

The polarization insensitive behavior of the MMI 3-dB coupler is crucial since polarization splitting is achieved after the 3-dB coupler by means of polarization selective photo-detector pairs.

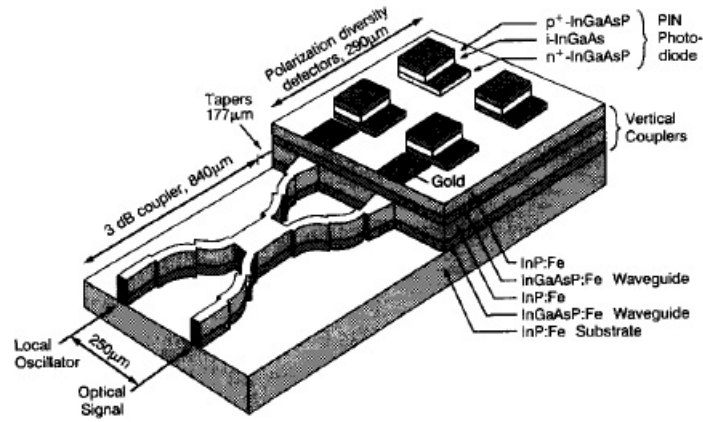


Figure 7.1: Coherent Receiver Front-end Schematic

This configuration has the advantage, that in addition to being suitable for polarization-diversity reception, it is also suitable for phase-diversity detection. Finally, the wavelength insensitive behavior of the 3-dB coupler in combination with the compact design of the photodetectors should result in a broad spectral operating range.

7.2 MZI

MZI [4] have been extensively used in practical realizations of optical processing because of their natural physical separation between the splitting/recombining functions and the phase-shifting function.

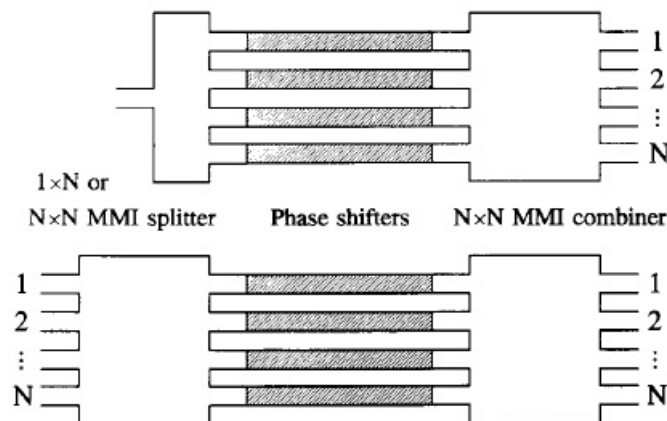


Figure 7.2: Schematic Layouts of a MZI

The extinction ratio in a MZI is directly limited by the imbalance of the input splitter and the output combiner. For example, a 0.2-dB power imbalance would limit the extinction ratio to -33 dB. The good balancing and stable relative phases shown by MMI couplers around their optimum operating point, together with their polarization insensitivity make these devices ideal candidates for integration into Mach-Zehnder structures (Figure 7.2). MZI with MMI splitters and recombiners have also been experimented in hollow dielectric waveguides.

The MMI couplers plays a crucial role in attaining large bandwidth and polarization independent operation. The possibility of achieving 1xN and NxN splitter/combiners with MMI devices allows the realization of very compact integrated multi-way optical switches.

7.3 Micro-ring Resonator

Microring resonators [35] have been used as a basic building block for optical signal processing applications such as optical switches, filters, modulators, and add-drop multiplexers. Almost all of the reported works on micro-ring resonator structures have used directional couplers as the coupling element. In order to meet a variety of requirements for high-speed signal processing, the coupling coefficient needs to be adjusted arbitrarily and the directional couplers can meet this requirement. However, for applications requiring high quality factor Q of the resonators, for high speed operation, the separations between two waveguides in the directional coupler must be very small. As a result, high loss due to conversion loss of modes is occurred recently. Also, the directional coupler has a large size and small fabrication tolerance.

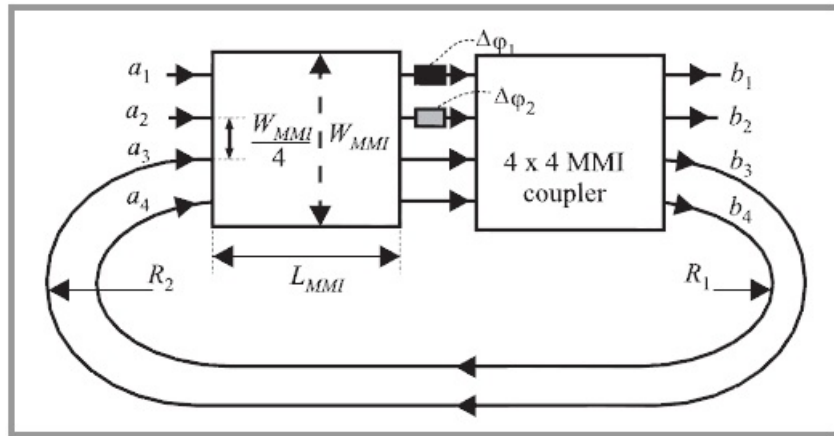


Figure 7.3: Structure of a Micro-ring Resonator

Therefore, MMI couplers are used in such structures (Figure 7.3) instead of directional couplers due to their advantages of compactness, ease of fabrication, large fabrication tolerance and ease of cascaded integration. A micro-ring resonator based on a 2x2 MMI coupler was demonstrated on SOI channel waveguides for the first time.

7.4 Ring Lasers

MMI devices have also proven to be successful out-coupling elements in ring lasers [4]. The out-coupling element has a crucial influence on the performance of the ring laser, since the out-coupler forms an integral part of the ring resonator.

The most commonly used out-coupling element is the Y-junction, which offers ease of design, but can only couple one of the counter-propagating beams out of the ring. Directional couplers have not proven to be a successful out-coupler, partly due to their incompatibility with high-contrast waveguides which are required to create low-loss small-radii bends. MMI devices on the other hand provide symmetric out-coupling, relaxed fabrication tolerances, ease of design and compatibility with high-contrast waveguides. An additional MMI combiner has been used to combine both counter-propagating beams in a single output waveguide, thereby increasing the efficiency.

Comparing the performance of ring lasers employing MMI couplers to those using Y-junctions, or directional couplers, the stability of the splitting ratio of the out-coupler is found to be the key factor. Lasers are subject to varying operating conditions; changes in current affect the gain spectrum, change the temperature and cause carrier induced refractive index changes. If the splitting ratio of the out-coupler, in turn, depends on wavelength, refractive index or excitation conditions, unstable lasing operation results, since laser performance is determined by the interplay of all these parameters. It is therefore the extreme stability of the splitting ratio of the MMI coupler, that leads to the improved performance of ring lasers incorporating MMI couplers.

Chapter 8

Conclusions and Future Prospects

Through the development of this project, we have reviewed the MultiMode Interference Coupler. The structure of this passive device is based on the MultiMode waveguide, and there for the theory that backs these structures was fully outlined. In addition, the self-imaging principle and its properties, are the functioning principals in which all MMI devices are grounded on.

It was proposed to design a MMI Coupler on a SOI platform, which lead to investigate its theoretical background, and to research about the several simulation tools based on numerical analysis, so the testing of the design would be accurate and reliable. The answer was the use of the most commercial photonic simulator developed to analyze general dielectric waveguides, the BeamPROP, which included the effects of both guided and radiating fields.

Through the use of this BPM based mode solver we performed simulations which permitted, not only the design but the optimization of the device, by determining the cause for the polarization dependence, the power loss and the imbalance of the outputs. The results where 4 couplers (1x3, 1x2, 2x2_{split}, and 2x2_{switch}) with a exponentially and parabolically, depending on the case, width tapered MMI section (when necessary) and linearly tapered access waveguide, that would eliminate, or lower the gravity of the mentioned problems, enhancing their performance.

To corroborate these attributes the fabrication tolerance of the width and length of the TE and TM polarizations, respectively, has been shown together with other parameters like the PDL and the EL, not only proving the success of the project, but demonstrating its advantages and qualities compared with other similar devices, so that it is clear that the MMI coupler the best option in several optical circuit applications.

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